ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Submarine Combat Systems Engineering Coordinator/Officer (CSEC/O)

Weapons Engineering Technician Occupation (W Eng Tech) (00366)
Naval Combat Systems Engineers (NCS Eng) (00344)

W Eng Tech – (AILN)
NCS Eng – (AILN)

MITE Course ID: – 121048

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Canadian Submarine Force and the Commandant of the Curriculum Control Authority (CCA) (Canadian Forces Naval Operations School (CFNOS)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the changes must be amplified in Annex J “Course History”.

Revision	Approval Date	Authority
1.0	16 Jun 07	DMTE
1.1	08 Dec 09	CFNOS
1.2	29 Nov 16	CCSF
1.3	9 Feb 22	NTDC(A)

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FOREWORD

1. This training document is developed under the authority of the Commander Canadian Submarine Force (CCSF) and IAW the following references:

   a. A-P9 - 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated 23 Apr 14;

   c. A-PD-055-002/PP-004 Non-Commissioned Member General Specification dated 09 Feb 11;

   d. A-PD-055-002/PP-002 W Eng Tech (00366) Job Based Specification dated 01 Sep 2011;

   e. A-PD-055-002/PP-001 Naval Combat Systems Engineering (NCS Eng) Officer (00344) Job Based Occupational Specification dated 10 May 10; and

   f. Naval Order (NAVORD) 4500-0 - Structured Delegation, and Naval Personnel and Training Order (NPTORD) 4500-22 - Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated Training Establishment (TE) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(A)	ALL	121048	AILN

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER CANADIAN SUBMARINE FORCE (CCSF)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Submarine Combat Systems Engineering Coordinator/Officer (CSEC/O). This QSP is designed to assist Training Establishments (TEs) in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated IAW NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a period of instruction is 50 minutes in duration.

	Periods
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	226
Enabling Checks (ECs)	53.5
Performance Checks (PCs)	29
Sub-total	**308.5**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	10
Training Administrative Time (TAT)	10
Training Travel Time (TTT)	0
Physical Training (PT)	24
Ceremonial/ Divisions	2
Sub-total	**46**
	354.5
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) 8 (50 min) periods * **Partial days must be rounded up**	**354.5/8= 45 trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Course Officer / Instructor	LCdr/Lt(N)	- CSEC/O (AILN) - Basic Submarine Qualification (BSQ) (AILS)	1	All

		- Naval Combat Systems Engineer (NCS Eng) Head-of-Department (HOD) Qualified (AISF) - Post-HOD tour		
Senior Instructor	CPO2/PO1	- BSQ (AILS) - CSEC/O (AILN) - Post-CSEC tour	1	All
Instructor	MS/PO2	- BSQ (AILS) - Weapons Engineering (W Eng) Submarine Fire Control Tech (AKQN)	1	EO 006.05
Instructor	MS/PO2	- Submarine W Eng Technician Sonar (AKQE) - BSQ (AILS)	1	EO 006.04 EO 006.03 EO 006.02
Instructor	MS/PO2	- W Eng Communications Maintainer Submarine (AKQK) - BSQ (AILS)	1	EO 006.01 EO 006.02 EO 006.03
Guest Speaker	Lt(N) & Above	- Submarine Operations Room Officer (SMORO) (AILU)	1	EO 004.03 Weapons Employment
Guest Speaker	N/A	- N37 Representative	1	PO 003
Guest Speaker	N/A	- Staff Officer of Fleet Technical Authority (FTA) Submarines	1	EO 003.01
Guest Speaker	Lt(N) & Above	- Sea Training (Submarines) (ST(S)) Staff	1	EO 003.02

Critical Training Resources	Time Required
Classroom	42 days
Secure Discussion Area	1 day
S-100 Weapons Handling & Distribution System Trainer (WHDST)	61 periods
Tactical Weapons System (TWS) Trainer	3 periods
S-17 Fred Sherwood Trainer	26 periods

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 - Quantity Control Process for the Naval Training System (NTS).

a. Maximum - 6;

b. Minimum - 1; and

c. Optimum - 3.

MITE

5. Loading. The authority to load members onto this course lies with the TE. CNPTG and CCSF are to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the TE conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. Security Clearance Level II (or higher);

b. (Maritime Command) MARCOM Warship Radiation Hazard (RADHAZ) Officer (AJCQ) qualification;

c. Non-Commissioned Members (NCMs):

(i) minimum rank PO2 W ENG Technician (00366);

(ii) Magazine Custodian course (MITE Crse ID 118948) qualified; and

(iii) BSQ Part 1 (MITE Crse ID 109201) qualified.

d. Officers:

(i) NCS ENG Basic Occupation Qualification (AIRX) (PH VI); and

(ii) BSQ Part 1 (MITE Crse ID 109201) qualified.

8. Qualification. Successful completion of this course constitutes the primary means of attaining the Combat Systems Engineering Coordinator/Officer (AILN), which is a Rank Qualification (RQ) for NCMs.

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW Chief of Military Personnel (CMP) Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted IAW NPTORD 4500-27 - PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. Recommended sequence of instruction, IAW Annex G to be PO 002, PO 003, PO 006, PO 001, PO 004, PO 005; and

14. Recommend additional Headquarters 1 (HQ1) support manning, with associated qualifications, as follows:

Support Manning	MS & Above	- BSQ (AILS) - Space 2 I/C Engineering Officer of the Watch (AICH) - Victoria Class Machinery Control Console (MCC) / Ship Control Console (SCC) Watchkeeper (AIND)	1	HQ1
Support Manning	MS & Above	- BSQ (AILS) - Victoria Class MCC/SCC Watchkeeper (AIND)	1	HQ1

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED
COMMANDER CANADIAN SUBMARINE FORCE (CCSF)

PO 001

1. Performance. Coordinate Unit Hazardous Material (HAZMAT) Responsibilities

2. Conditions

   a. Given:

      (1) Department of National Defense (DND) workstation/laptop;
      (2) Hazardous material spill response equipment;
      (3) HAZMAT related software;
      (4) Access to outside agencies;
      (5) References; and
      (6) Assistance.

   b. Denied: Supervision.

   c. Environmental: HMC Submarine.

3. Standard. The CSEC/O shall (B0900) coordinate unit HAZMAT responsibilities, IAW references, by:

   a. (B0910) Managing unit HAZMAT program readiness; and

   b. (B0890) Directing response to a HAZMAT incident.

4. References. A45, A53, A54, A55, A90, A91, A92, A93, C1, C2, C3.

5. Task Numbers. B0740, B0180, B0040, B0350, B0890, B0900, B0910, and B0930.

6. Training Limitations. Nil.

PO 002

1. Performance. Maintain Divisional Administrative Readiness

2. Conditions

   a. Given:

      (1) DND workstation/laptop;
      (2) Administrative software;
      (3) Access to outside agencies; and
      (4) References.

   b. Denied: Supervision.

   c. Environmental:

      (1) HMC Submarines; and
      (2) Shore establishments.

3. Standard. The CSEC/O shall maintain divisional administrative readiness, by:

   a. (B0920) Managing Combat Systems Engineering Department (CSED) safety IAW Safety & Environment Management System (SEMS); and

   b. (R1140) Managing CSED personnel readiness.

4. References. A1, A3, A86, A89, A90, A91, A92.

5. Task Numbers. B0250, B0100, B0030, B0120, R0310, R0730, A0840, R0750, R0010, R1040, R0330, A0470, B0920, and R1140.

6. Training Limitations. Recommend EO 002.01 & EO 002.02 for Distance Learning (DL) development (17 periods).

PO 003

1. Performance. Coordinate Divisional Technical Readiness

2. Conditions

   a. Given:

      (1) DND workstation/laptop, with standalone Message Drafter software;
      (2) Submarine Combat Systems (CS) suite equipment;
      (3) Defense Resource Management Information System (DRMIS) or equivalent;
      (4) Access to outside agencies;
      (5) References; and
      (6) Assistance.

   b. Denied: Supervision.

   c. Environmental:

      (1) HMC Submarines; and
      (2) Shore establishments.

3. Standard. The CSEC/O shall (A1210) coordinate divisional technical readiness, IAW references, by:

   a. (A1220) Managing technical administration;

   b. Contributing to the project management process;

   c. (A1040) Coordinating unit-level maintenance activities, to include:

      (1) Managing unit-level maintenance;

      (2) (A1240) Directing CSE corrective maintenance (CM);

      (3) (A1250) Coordinating system trials; and

      (4) (A1230) Directing Combat Systems Engineering (CSE) preventative maintenance (PM).

   d. (A0450) Managing CSED pre-sail checks; and

   e. (S1610) Supporting towed-array operations.

4. References. A4, A6, A7, A9, A10, A53, A54, A55, A58, A59, A60, A61, A62, A82, A83, A84, A85, A87, A90, A92, A93, A94, C6.

5. Task Numbers. A1260, S1330, S1320, A0640, A1040, A1230, A0830, A1000, A0120, A0040, A1240, A1270, A0870, A0190, A1280, A0990, O0890, A1290, A1110,

A0860, A1300, O0320, A0450, A0210, A1160, A1200, A1170, S1610, S1340, F1260, F1250, F1130, F1120, F1230, F1240, H1070, H1110, H1100, L1020, L1010, L0950, L0940, N0680, S1510, H0460, H0970, H0980, N1090, N0610, N0620, N1080, N1070, H0450, H0990, I0960, I0330, K1020, K1060, K1050, K1080, F1200, F1100, F1150, F1140, F1190, M1120, M1050, M1110, H1120, H0010, H1060, H1080, H1090, O0310, O0990, O1000, S1500, N1100, I0970, P0290, P0880, I0340, M1130,M1060, M1140, H1000, H0060, H1010, H1020, N1010, N0670, N1020, N1120, N1050, N1110, N1180, N1170, F1280, F1270, F1210, F1110, F1220, H0130, Q0100, Q0080, Q0880, Q0930, Q0920, M1030, M1070, M1080, M0710, M1150, M1160, S1590, S1600, M1170, M0870, M0880, O0870, K0970, I0840, J1150, K1030, K1040, K0910, J0110, J1090, J0120, K0930, J1160, K0950, K0920, K1000, K1010, K0960, K0880, J1140, J0390, J1260, J0400, J1250, J0150, J0160, J1100, J1170, J1180, J0330, J0340, J1230, J1240, A1250, S1660, J1130, E1290, E1300, E1310, E1320, E1330, G1390, G1400, G1410, G1420, G1430, G1440, H1140, H1150, H1160, H1170, H1180, H1200, H1210, H1220, H1230, H1240, H1260, H1270, H1280, H1290, H1300, H1310, I0990, I1000, I1010, I1020, I1030, I1050, I1060, I1070, I1080, I1090, I1110, I1120, I1130, I1140, I1150, K1080, K1090, K1100, K1110, K1120, L1060, L1070, L1080, L1090, L1100, M1220, M1230, M1240, M1250, M1260, N1220, N1230, N1240, N1250, N1260, H1370, H1380, H1390, S1690, S1700, S1710, S1730, S1740, S1750, I1160, I1170, I1180, L0990, L1000, L0660, L0680, L0860, L0870, M0800, M0810, M0360, M0390, F0090, F0120, F0130, F0140, M0700, Q0260, Q0270, I0850, O0910, O0920, O0630, O0640, I0910, A1210, A1220, K0180, P0300, P0960, P0970, G1450, G1460, G1470, G1480, G1490, H1320, H1330, H1340, H1350, and H1360.

6. Training Limitations. Recommend this PO for DL development (54 periods).

PO 004

1. Performance. Support the Weapons Certification Programme (WCP)

2. Conditions

   a. Given:

      (1) Equipment, IAW Submarine Weapons Operating Procedures (SUBWOPS);
      (2) DND workstation/laptop;
      (3) DRMIS or equivalent;
      (4) Access to outside agencies;
      (5) References; and
      (6) Assistance.

   b. Denied: Supervision.

   c. Environmental:

      (1) HMC Submarines; and
      (2) Shore establishments.

3. Standard. The CSEC/O shall (A1310) support the WCP, IAW references, by:

   a. (A1320) Supporting WCP administration;

   b. (S1620) Supporting Weapons Handling & Discharge System (WHDS) training; and

   c. (S1630) Supporting weapons employment for WCP.

4. References. A4, A8, A45, A46, A47, A49, A53, A54, A80, A81, A86, A88.

5. Task Numbers. A1310, A1320, S1310, F0150, B0270, S1640, F0020, F0760, F0740, F0040, F0030, S1620, S1630, F0880.

6. Training Limitations. Availability of trainers.

PO 005

1. Performance. Conduct duties of CS Engineering Subject Matter Expert (SME) in HQ1.

2. Conditions

a. Given:

(1) CS-suite;
(2) Emergency Response Team (ERT); and
(3) References.

b. Denied:

(1) Assistance; and
(2) Supervision.

c. Environmental: HMC Submarine.

3. Standard. The CSEC/O shall (S1650) conduct the duties of CS Engineering SME in HQ1, IAW references, by sustaining combat systems in emergency situations, to include:

a. (A1080) Diagnosing CS suite-level faults;

b. (B0860) Directing an ERT; and

c. (A0620) Providing recommendations to Command on CS equipment availability and reconfiguration options.

4. References. A72, A73, A74, A75, A76, A77, A78, A79, A91, A92, A93.

5. Task Numbers. S1650, A1080, B0860, A0620, and I0230.

6. Training Limitations. Nil.

PO 006

1. Performance. Advise Command on the CS suite.

2. Conditions

a. Given:

(1) References;
(2) DND workstation/laptop; and
(3) Submarine CS-suite equipment.

b. Denied: Assistance.

c. Environmental:

(1) HMC Submarine; and
(2) Shore establishment.

3. Standard. The CSEC/O shall (A1330) advise Command on the CS suite, IAW references, by:

a. Advising Command on the communications suite;

b. Advising Command on the electronic warfare (EW) suite;

c. Advising Command on the navigation suite;

d. Advising Command on the sonar suite;

e. Advising Command on the fire control suite; and

f. Advising Command on the CS suite integration.

4. References. A2, A5, A8, A11, A12, A13, A14, A15, A16, A18, A19, A20, A21, A22, A23, A24, A25, A26, A27, A28, A29, A30, A31, A32, A33, A34, A35, A36, A37, A38, A39, A40, A41, A42, A43, A44, A48, A49, A50, A51, A52, A53, A54, A55, A56, A57,A63, A64, A65, A66, A67, A68, A69, A70, A71, A79 C4, C5.

5. Task Numbers. O0170, M0150, M0160, M0130, M0180, N0130, N0140, N0170, F0250, H0190, S1430, H0220, H0230, Q0150, L0140, L0150, J0180, J0170, J0220, J0190, J0210, E1280, K0210, P0180, I0160, H0210, F0260, F0270, F0280, S1670, A1330, G1380, G1440, H1130, H1190, H1250, H1310, I0980, I1040, I1100, K1070, L1050, M1210, N1210, S1680, S1720, I0150, AND O0140.

6. Training Limitations. Recommend this PO for DL development (81 periods).

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA (CFNOS)

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (PRACTICAL or WRITTEN) examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative assessments are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. The Training Development Centers (TDCs) provide quality control by designing, administering and scoring all PCs. Any PC not administered or invigilated by TDC staff will be administered by a non-involved instructor designated by the senior instructor. ECs are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional / TDC staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programs and TE Standing Orders (SO) or Standard Operating Procedures (SOP).

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the Training Review Board (TRB) or Progress Review Board (PRB) (if required); and

d. Information to assist in completion of the Course Report (CF 377).

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW TE SOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW Defense Administrative Orders and Directives (DAOD) 5031-09 & NPTORD 4500-51 - Course Reporting and Certificates.

12. Trainee will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

13. Where necessary for RQ, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and / or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the TE.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT OF THE CCA (CCSF)

PO 001 Coordinate Unit Hazardous Material (HAZMAT) Responsibilities

EO 001.01 Manage Unit HAZMAT Programme Readiness

1. Performance. Manage unit HAZMAT programme readiness.

2. Conditions

   a. Given:

      (1) References;
      (2) Laptop / workstation;
      (3) Assistance; and
      (4) Supervision.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall (B0910) manage unit HAZMAT programme readiness, IAW references, by:

   a. (B0740) Conducting a HAZMAT inspection;

   b. (B0930) Directing the maintenance of equipment associated with HAZMAT clean up; and

   c. (B0180) Enforcing HAZMAT storage compatibility regulations.

4. Time Required

   a. EO - 3 x 50 min periods;

   b. EC - 0.5 x 50 min period (review / de-brief); and

   c. PC - 6 x 50 min periods (1 period / trainee).

5. Instructional Methods

   a. Guided discussion; and

   b. Self-study.

6. Training Aids. Nil.

7. Test Details. EO 001.01 will be assessed in EC 001.01 as a take-home assignment, and in PC 001.

8. References. See para 10.

9. Remarks. Recommend this EO for DL development.

10. Teaching Points (TP)

	Description	Instructional Method	Time	Reference
TP1	(B0740) Conduct a HAZMAT inspection - Policies - Outside agencies - Reports	Guided discussion Self-study	1 x 50 mins	A53, A54, A55, A90, A91, A92, A93, C1, C2, C3
TP2	(B0930) Direct the maintenance equipment associated with HAZMAT clean up - Equipment - Certification		1 x 50 mins	
TP3	(B0180) Enforce HAZMAT storage compatibility regulations		1 x 50 mins	

Enabling Check - EO 001.01 - Manage Unit HAZMAT Programme Readiness

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.01 will be a take-home written assessment based on content from EO 001.01.

3. Pass mark will be 70%.

4. Time required: 0.5 x 50 min period to review / de-brief take-home assignment.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a take-home written assessment based on content from EO 001.01.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

3. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

4. Read each question carefully before answering.

5. If you have any questions during this assessment, raise your hand.

6. References will be provided.

7. Pass mark is 70%.

8. We will have 0.5 x 50 min period to review / de-brief this take-home assignment.

9. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Take-home assignment to include a minimum of ten short-answer questions, where trainees must cite references used.

PART IV – ENABLING CHECK LIST

Assessment Item	Percentage
TP1 – Conduct a HAZMAT inspection	40
TP2 – Direct the maintenance equipment associated with HAZMAT clean-up	40
TP3 - Enforce HAZMAT storage compatibility regulations	20

The trainee must achieve 70% to pass this EC.

PO 001 Coordinate Unit HAZMAT Responsibilities

EO 001.02 Direct Response to a HAZMAT Incident

1. Performance. Direct response to a HAZMAT incident.

2. Conditions

   a. Given:

      (1) HAZMAT Clean-up Team;
      (2) Otto Fuel Monitor (OFM);
      (3) Mark (Mk) 48 spill simulator;
      (4) Hazardous material spill response equipment ; and
      (5) References.

   b. Denied:

      (1) Assistance; and
      (2) Supervision.

   c. Environmental:

      (1) Classroom; and
      (2) WHDST.

3. Standard. The trainee shall (B0890) direct response to a HAZMAT incident, IAW references, by:

   a. (B0350) Operating HAZMAT detection equipment;

   b. (B0040) Classifying/identifying HAZMAT; and

   c. Responding to a HAZMAT incident.

4. Time Required

   a. EO - 9 x 50 min periods; and

   b. EC - 6 x 50 min periods (1 period / trainee).

5. Instructional Methods

   a. Demonstration - performance;

   b. Guided discussion;

   c. Study assignment; and

   d. Simulation.

6. Training Aids. Nil.

7. Test Details. EO 001.02 will be assessed in EC 001.02 and PC 001.

8. References. See para 10.

9. Remarks

   a. TP1 - OFM should be powered up 24 hours prior (self-test);

   b. Mk 48 spill simulator should be powered up and tested 24 hours prior; and

   c. TP3 - During the practical assessment, trainees will participate as HAZMAT Clean-up team in secondary roles.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	(B0350) Operate HAZMAT detection equipment - Operate OFM - Self-diagnostic tests	Demonstration - performance	1 x 50 mins	A45, A53, A54, A55, A90, A91
TP2	(B0040) Classify/identify HAZMAT - Utilizing references - First aid response	Demonstration - performance Study assignment Guided discussion	1 x 50 mins	C1, C2

TP3	Respond to a HAZMAT incident
	- Procedure, IAW fleet policy
	  ○ Contain
	  ○ Confine
	  ○ Casualty
	  ○ Clean-up
	  ○ Disposal
		Demonstration - performance
		
		Simulation
			1 x 50 mins
			
			6 x 50 mins (1 period / trainee)
				A53, A54, A55, A90, A91, A92, A93, C1, C2, C3

Enabling Check - EO 001.02 - Direct Response to a HAZMAT Incident

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. WHDST Trainer; and/or

2. Classroom.

Arrangements and Procedures

3. EC 001.02 will be a practical assessment based on content from EO 001.02.

4. Pass mark will be 5 of 7 Satisfactory (SAT).

5. Time required: 6 x 50 min periods (1 period / trainee).

6. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a practical assessment based on content from EO 001.02.

2. During this practical assessment, you will be required to direct a response to a HAZMAT incident. You must pass this assessment to pass EO 001.02.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical assessment will consist of trainee responding to an HAZMAT incident scenario.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	Unsatisfactory (UNSAT)	Comments
Operate HAZMAT detection equipment			
Classify/identify HAZMAT IAW references			

Performance Factors	SAT	Unsatisfactory (UNSAT)	Comments
Respond to a HAZMAT incident			
Contain			
Confine			
Casualty			
Clean-up			
Disposal			

The trainee must achieve 5 of 7 SAT to pass this EC.

PERFORMANCE CHECK - PO 001 - Coordinate Unit HAZMAT Responsibilities

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer; and/or

2. Classroom.

Arrangements and Procedures

3. PC 001 will be a practical assessment based on content from PO 001.

4. Pass mark will be 5 of 7 SAT.

5. Time required: 6 x 50 min periods (1 period / trainee).

6. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will be required to direct a response to a HAZMAT incident. You must pass this assessment to pass PO 001.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Practical assessment - Trainee responds to a HAZMAT incident scenario. This scenario must be different from that of EC 001.02.

PART IV – PERFORMANCE CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Operate HAZMAT detection equipment			
Classify/identify HAZMAT IAW references			
Respond to a HAZMAT incident			

Performance Factors	SAT	UNSAT	Comments
Contain			
Confine			
Casualty			
Clean-up			
Disposal			

The trainee must achieve 5 of 7 SAT to pass this PC.

PO 002 Maintain Divisional Administrative Readiness

EO 002.01 Manage CSED Safety IAW SEMS

1. Performance. Manage CSED Safety IAW SEMS.

2. Conditions

   a. Given:

      (1) References; and
      (2) Laptop / Workstation.

   b. Denied:

      (1) Assistance; and
      (2) Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall (B0920) manage CSED safety IAW SEMS, IAW references, by:

   a. Enforcing safety policies; and

   b. Enforcing shipboard safety.

4. Time Required

   a. EO - 7 x 50 min periods;

   b. EC - 1 x 50 min period; and

   c. PC - 3 x 50 min periods.

5. Instructional Methods

   a. Interactive lecture;

   b. Self-study; and

   c. Case study.

6. Training Aids

   a. Sample CF 98;

   b. Sample DND 663;

   c. Lockout / Tagout (LOTO) Devices; and

   d. Sample LOTO form.

7. Test Details. EO 002.01 will be assessed in EC 002.01 and PC 002.

8. References. See para 10.

9. Remarks

   a. Recommend this EO for DL development; and

   b. Self-study to include reviewing equipment handouts and reading publications in preparation for lessons.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Safety policies - Personnel - Equipment - Environmental	Interactive lecture Case study Self-study	2 x 50 mins	A1, A86, A90, A92
TP2	Shipboard Safety - Shipboard organization - Supervisor responsibilities - (B0120) Draft report on Injuries or Exposure to Toxic Material (CF 98) - (B0030) Investigate / Review a CF 98 - (B0100) Draft a General Safety Accident Prevention Report (DND 663) - (B0250) Investigate / Review a DND 663 - Lock Out/Tag Out - Personal Protection Equipment (PPE) (Fall Arrest, harnesses) - Shipboard safety mast control - Weapons firing safety organization and key control - Confined space entry & gas-free policies		5 x 50 mins	

Enabling Check - EO 002.01 - Manage CS Department Safety IAW SEMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 002.01 will be a written assessment based on content from EO 002.01.

3. Pass mark will be 70%.

4. Time required: 1 x 50 min period.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a written assessment based on content from EO 002.01.

2. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

6. Read each question carefully before answering.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 70%.

10. You will have 1 x 50 min period to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Written assessment to consist of a minimum of 10 short-answer questions (publication search), with references cited, focused on LOTO and including a question related to CF 98 / DND 663.

PART IV – ENABLING CHECK LIST

Assessment Item	Percentage
TP1 – Safety Policies	20
TP2 – Shipboard Safety	80

The trainee must achieve 70% to pass this EC.

PO 002 Maintain Divisional Administrative Readiness

EO 002.02 Manage CSED Personnel Readiness

1. Performance. Manage CSED Personnel Readiness.

2. Conditions

   a. Given:

      (1) References;
      (2) Laptop / Workstation; and
      (3) Assistance.

   b. Denied: Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall (R1140) manage CSED personnel readiness, IAW references, by:

   a. (A0470) Managing the CSED as an independent HOD or Chief-of-Department (CHOD);

   b. (R0310) Maintaining watch and station bill;

   c. (R0750) Producing a departmental annual training plan; and

   d. (R0330) Monitoring the training progress of personnel.

4. Time Required

   a. EO - 5 x 50 min periods; and

   b. EC - 1 x 50 min period.

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids. Sample watch and station bill.

7. Test Details. EO 002.02 will be assessed in EC 002.02 and in PC 002.

8. References. See para 10.

9. Remarks

a. Self-study to include reviewing equipment handouts and reading publications in order to prepare for lessons; and

b. Recommend this EO for DL development.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	(A0470) Manage the CSED as an independent HOD or CHOD	Interactive lecture Guided discussion Self-study	2 x 50 mins	A92
TP2	(R0310) Maintain watch and station bill - Check CSED manning IAW VCSSO - Check appropriate number of Control Room Watchkeeper (Dived) (CRWK(D)) qualified personnel on submarine - Check appropriate number of Control Room Watchkeeper Supervisor (CRWK(S)) qualified personnel on submarine - Check the balance of SMEs on watches		1 x 50 mins	A91, A92
TP3	(R0750) Produce a departmental annual training plan - (A0840) Track CS Combat Readiness Requirements (CRR) progress - (R0730) Plan CS collective / team training (e.g. ERT/HAZMAT exercises)		1 x 50 mins	A3, A91
TP4	(R0330) Monitor the training progress of personnel - (R0010) Establish a training priority - (R1040) Determine subordinates individual training and educational requirements - Maintain personnel training profiles		1 x 50 mins	A3, A89

Enabling Check - EO 002.02 - Manage CSED Personnel Readiness

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 002.02 will be a written assessment based on content from EO 002.02.

3. Pass mark will be 70%.

4. Time required: 1 x 50 min period to complete this assignment.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.02 will be a written assessment based on content from EO 002.02.

2. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

6. Read each question carefully before answering.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 70%.

10. You will have 1 x 50 min period to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. This EC will consist of at least ten short-answer questions with a focus on the watch and station bill.

PART IV – ENABLING CHECK LIST

Assessment Items	Percentage
TP1 - Manage the CSED as an independent HOD or CHOD	20
TP2 - Watch and station bill	40
TP3 - Departmental annual training plan	20
TP4 - Monitor the training progress of personnel	20

The trainee must achieve 70% to pass this EC.

PERFORMANCE CHECK - PO 002 - Maintain Divisional Administrative Readiness

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 002 will be a two-part practical assessment based on content from PO 002.

3. Pass mark will be 6 out of 8 SAT.

4. Time required: 3 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will be required to Maintain Divisional Administrative Readiness. You must pass this assessment to pass PO 002.

3. References will be provided.

4. Pass mark is 6 out of 8 SAT.

5. You will have 3 x 50 min periods to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

7. Two-part, scenario-based assessment:

   a. PC Part A. Trainees are provided with a list of CSED members, with positions and qualifications. Trainees must complete the CSE positions on a watch and station bill, adhering to submarine licensing, and must identify any problems WRT positions and qualifications. The product will be a complete watch and station bill IAW with PC List; and

   b. PC Part B. One CSED member on the watch and station bill becomes injured (injured personnel must impact at least four qualifications). Review the member's DND 663 and CF 98 (which will contain at least one error). Identify and explain the error(s) on the form. Identify the implications of this incident on the watch and station bill and provide recommendations to Command on how to fill the critical manning requirement.

PART IV – PERFORMANCE CHECK LIST

PC Part A

Performance Factors	SAT	UNSAT	Comments
Complete watch and station bill IAW VCSSO			
Correct number of CRWK(D) qualified personnel			
Correct number of CRWK(S) qualified personnel			
Balance of SMEs on watches			

PC Part B

Performance Factors	SAT	UNSAT	Comments
Identify all errors on DND 663			
Identify all errors on CF 98			
Brief impact on watch and station bill / submarine			
Provide recommendations to fill critical manning			

The trainee must achieve 6 out of 8 SATs on parts A & B to pass this PC.

PO 003 Coordinate Divisional Technical Readiness

EO 003.01 Manage Technical Administration

1. Performance. Manage technical administration.

2. Conditions

   a. Given:

      (1) References;

      (2) Laptop / Workstation with DRMIS; and

      (3) Technical administration forms/logs.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall (A1220) manage technical administration, IAW references, by:

   a. (A1260) Maintaining configuration management;

   b. (S1660) Conducting the duties of the Small Arms Custodian;

   c. Applying Submarine Safety (SUBSAFE) policy; and

   d. Managing the Tiered Readiness Program (TRP).

4. Time Required

   a. EO - 19 x 50 min periods;

   b. EC - 1 x 50 min period; and

   c. PC - 1 x 50 min period.

5. Instructional Methods

   a. Interactive/guest lecture; and

   b. Case study.

6. Training Aids

   a. Form templates:

      (1) Periodic Engineering Letter (PEL);

      (2) Engineering Change (EC); and

(3) Unsatisfactory Condition Report (UCR).

b. Defence Wide Area Network (DWAN) capable workstation / laptop;

c. Small arms log; and

d. Audio / visual aids.

7. Test Details. EO 003.01 will be assessed in EC 003.01 and PC 003.

8. References. See para 10.

9. Remarks

a. TP 1 – Guest lecture from N37-1;

b. TP 3 – Guest lecture from FTA; and

c. Recommend this EO for DL development.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	(A1260) Maintain configuration management - (A0640) Provide input to a Periodic Engineering Letter (PEL) - Engineering Change (EC) - Unsatisfactory Condition Report (UCR)	Interactive / Guest lecture	3 x 50 mins	A6, A7
TP2	(S1660) Small arms custodian - Duties/responsibilities o Lockers o Maintenance o Logs o Rounds o Security o Safety - Regulations/policies	Interactive lecture	2 x 50 mins	A82, A83, A85, A92
TP3	SUBSAFE Policy - Overview - Authorities - Responsibilities - Principles	Interactive / Guest lecture	12 x 50 mins	A84

	- License types
	- License validity
	- Functional pillars
	  ○ Material Certification (MATCERT)
	- (A1280) Conduct risk assessment
	  ○ Case studies
	- Policy compliance exceptions
		Case study		

TP4	TRP - Overview - Procedure / Managing - Timelines - Work-ups (WUPS) - Collective team and refresher training - Technical trials	Interactive lecture	2 x 50 mins	A4, A94

Enabling Check - EO 003.01 - Manage Technical Administration

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom;

2. References; and

3. Workstation / Laptop.

Arrangements and Procedures

4. EC 003.01 will be a written assessment based on content from EO 003.01.

5. Pass mark will be 70%.

6. Time required: 1 x 50 min period.

7. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a written assessment based on content from EO 003.01.

2. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

6. Read each question carefully before answering.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 70%.

10. You will have 1 x 50 min period to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. EC will consist of a minimum of ten short-answer questions, covering all TPs. Emphasis on SUBSAFE policy, especially in the areas of authorities, license types and validity, Material Certification, and risk assessment. References must be cited with answers.

PART IV – ENABLING CHECK LIST

Assessment Items	Percentage
TP1 – Configuration management	10
TP2 – Small arms custodian	20
TP3 – SUBSAFE Policy	40
TP4 – TRP	30

The trainee must achieve 70% to pass this EC.

PO 003 Coordinate Divisional Technical Readiness

EO 003.02 Coordinate Unit-Level Maintenance Activities

1. Performance. Coordinate unit-level maintenance activities.

2. Conditions

   a. Given:

      (1) Laptop / Workstation, with standalone Message Drafter software and/or DRMIS;

      (2) References; and

      (3) Assistance.

   b. Denied: Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall (A1040) coordinate unit-level maintenance activities, IAW references, by:

   a. Managing unit-level maintenance;

   b. (A1230) Directing CM;

   c. (A1250) Coordinating system trials; and

   d. (A1240) Directing PM.

4. Time Required

   a. EO - 15 x 50 min periods; and

   b. EC - 1 x 50 min period (review / debrief).

5. Instructional Methods

   a. Interactive/guest lectures;

   b. Guided discussion;

   c. Self-study; and

   d. Case study.

6. Training Aids

   a. Sample TRP OPORD;

   b. Sample Operational Deficiency (OPDEF); and

c. Audio/visual aids.

7. Test Details. EO 003.02 will be assessed in EC 003.02 as a self-study assignment, and in PC 003.

8. References. See para 10.

9. Remarks

a. The maintenance procedure, in TP1, applies to all CS equipment, as per this QSP Annex B, Task List, containing all components deemed as trained tasks for this job. The following tasks, although not specifically mentioned in the Teaching Points of this EO, are trained tasks on the job for CSEC/O's, and follow the maintenance procedure as outlined in TP1:

L0660	Analyze/evaluate/interpret a Message Processing and Handling System functional check result
L0680	Analyze/evaluate/interpret a Message Processing and Handling System test/trial/inspection result or report
L0940	Direct a corrective maintenance/repair activity on a Message Processing and Handling System
L0990	Direct a Message Processing and Handling System functional check
L1000	Direct a Message Processing and Handling System PM activity (including a test/trial/inspection)
L0860	Analyze/evaluate/interpret a Red/Black Message Sub System functional check result
L0870	Analyze/evaluate/interpret a Red/Black Message Sub System test/trial/inspection result or report
L0950	Direct a corrective maintenance/repair activity on a Red/Black Message Sub System
L1010	Direct a Red/Black Message Sub System functional check
L1020	Direct a Red/Black Message Sub System PM activity (including a test/trial/inspection)
H0450	Analyze/evaluate/interpret an UWT System test/trial/inspection result or report
H0460	Analyze/evaluate/interpret an UWT Systems functional check result
H0970	Direct a corrective maintenance/repair activity on an UWT System
H0980	Direct an UWT System PM activity (including a test/trial/inspection)
H0990	Direct an UWT System functional check
S1500	Analyze/evaluate/interpret a Tactical Television System functional check result
S1510	Direct a corrective maintenance/repair activity on a Tactical Television System
S1690	Direct a Tactical Television System functional check result
S1700	Direct a Tactical Television System preventive maintenance
S1710	Analyze a Tactical Television System test/trial/inspection result or report
N0610	Analyze/evaluate/interpret a Depth Sounder System functional check result
N0620	Analyze/evaluate/interpret a Depth Sounder System test/trial/inspection result or report
N1010	Direct a corrective maintenance/repair activity on a Depth Sounder System
N1070	Direct a Depth Sounder System functional check
N1080	Direct a Depth Sounder System PM activity (including a test/trial/inspection)

N0670	Analyze/evaluate/interpret a Doppler Speed Log System functional check result
N0680	Analyze/evaluate/interpret a Doppler Speed Log System test/trial/inspection result or report
N1020	Direct a corrective maintenance/repair activity on a Doppler Speed Log System
N1090	Direct a Doppler Speed Log System functional check
N1100	Direct a Doppler Speed Log System PM activity (including a test/trial/inspection)
I0330	Analyze/evaluate/interpret an Internal Communication System functional check result
I0340	Analyze/evaluate/interpret an Internal Communication System test/trial/inspection result or report
I0910	Direct a corrective maintenance/repair activity on an Internal Communication System
I0960	Direct an Internal Communication System functional check
I0970	Direct an Internal Communication System PM activity (including a test/trial/inspection)
F1100	Direct a corrective maintenance/repair activity on a Submarine Fire Control System (SFCS)
F1140	Analyze/evaluate/interpret a SFCS functional check result
F1150	Analyze/evaluate/interpret a SFCS test/trial/inspection result or report
F1190	Direct a SFCS functional check
F1200	Direct a SFCS PM activity (including a test/trial/inspection)
K0180	Analyze/evaluate/interpret a Wide Access (WAN) / Local Access Network (LAN) System test/trial/inspection result or report
K0970	Direct a corrective maintenance/repair activity on a WAN/LAN System
K1020	Analyse WAN/LAN event logs and reports
K1050	Direct a WAN/LAN System functional check
K1060	Direct a WAN/LAN System PM activity (including a test/trial/inspection)
M0800	Analyze/evaluate/interpret an Automatic Information System (AIS) functional check result
M0810	Analyze/evaluate/interpret an AIS System test/trial/inspection result or report
M1050	Direct a corrective maintenance/repair activity on an AIS System
M1110	Direct an AIS System functional check
M1120	Direct an AIS System PM activity (including a test/trial/inspection)
H1010	Direct a Bathy System PM activity (including a test/trial/inspection)
H1060	Direct a corrective maintenance/repair activity on a Submarine Weapon System Data Bus (WSDB)
H1080	Direct a Submarine WSDB functional check
H1090	Direct a Submarine WSDB PM activity (including a test/trial/inspection)
H1120	Analyze/evaluate/interpret a Submarine WSDB functional check result
O0310	Analyze/evaluate/interpret an Interrogator Friend-or-Foe (IFF) System functional check result
O0320	Analyze/evaluate/interpret an IFF System test/trial/inspection result or report
O0890	Direct a corrective maintenance/repair activity on an IFF System
O0990	Direct an IFF System functional check

O1000	Direct an IFF PM activity (including a test/trial/inspection)
M1130	Direct an Internal Navigation System functional check
M0360	Analyze/evaluate/interpret an Internal Navigation System functional check result
M0390	Analyze/evaluate/interpret an Internal Navigation System test/trial/inspection result or report
M1060	Direct a corrective maintenance/repair activity on an Internal Navigation System
M1140	Direct an Internal Navigation System PM activity (including a test/trial/inspection)
H0060	Analyze/evaluate/interpret a Bathy System functional check result
H1000	Direct a Bathy System functional check
H1010	Direct a Bathy System PM activity (including a test/trial/inspection)
H1020	Direct a corrective maintenance/repair activity on a Bathy System
H0130	Analyze/evaluate/interpret a Bathy System test/trial/inspection result or report
N1050	Direct a corrective maintenance/repair activity on a SHINADS system
N1110	Analyze/evaluate/interpret a SHINADS functional check result
N1120	Analyze/evaluate/interpret a SHINADS system test/trial/inspection result or report
N1170	Direct a SHINADS functional check
N1180	Direct a SHINADS system PM activity (including a test/trial/inspection)
F1110	Direct a corrective maintenance/repair activity on a Submarine Periscope System
F1210	Direct a Submarine Periscope System functional check
F1220	Direct a Submarine Periscope System PM activity (including a test/trial/inspection)
F1270	Analyze/evaluate/interpret a Submarine Periscope System functional check result
F1280	Analyze/evaluate/interpret a Submarine Periscope System test/trial/inspection result or report
Q0080	Analyze/evaluate/interpret a Training Simulator functional check result
Q0100	Analyze/evaluate/interpret a Training Simulator test/trial/inspection result or report
Q0880	Direct a corrective maintenance/repair activity on a Training Simulator
Q0920	Direct a Training Simulator functional check
Q0930	Direct a Training Simulator PM activity (including a test/trial/inspection)
F0090	Analyze/evaluate/interpret a Torpedo Handling System functional check result
F0120	Analyze/evaluate/interpret a Torpedo Handling System test/trial/inspection result or report
F1120	Direct a corrective maintenance/repair activity on a Torpedo Handling System
F1230	Direct a Torpedo Handling System functional check
F1240	Direct a Torpedo Handling System PM activity (including a test/trial/inspection)
Q0260	Analyze/evaluate/interpret an Emergency Beacon functional check result
Q0270	Analyze/evaluate/interpret an Emergency Beacon system test/trial/inspection result or report

I0840	Analyze/evaluate/interpret a Submarine Escape Communication System functional check result
I0850	Analyze/evaluate/interpret a Submarine Escape Communication System test/trial/inspection result or report
I1160	Direct a Submarine Escape Communication System functional check result
I1170	Direct a Submarine Escape Communication System corrective maintenance
I1180	Direct a Submarine Escape Communication System preventive maintenance
F0130	Analyze/evaluate/interpret a Torpedo Launching System functional check result
F0140	Analyze/evaluate/interpret a Torpedo Launching System test/trial/inspection result or report
F1130	Direct a corrective maintenance/repair activity on a Torpedo Launching System
F1250	Direct a Torpedo Launching System functional check
F1260	Direct a Torpedo Launching System PM activity (including a test/trial/inspection)
P0290	Analyze/evaluate/interpret an Event Recorder System functional check result
P0300	Analyze an Event Recorder System test/trial/inspection result or report
P0880	Direct a corrective maintenance/repair activity on an Event Recorder System
P0960	Direct an Event Recorder System functional check
P0970	Direct an Event Recorder System PM activity (including a test/trial/inspection)
S1590	Analyze/evaluate/interpret a Desiccant System functional check result
S1600	Direct a corrective maintenance/repair activity on a Desiccant System
S1730	Direct a Desiccant System functional check result
S1740	Direct a Desiccant System preventive maintenance
S1750	Analyze a Desiccant System test/trial/inspection result or report
I0990	Direct a VICES functional check
I1000	Direct a VICES corrective maintenance
I1010	Direct a VICES preventive maintenance
I1020	Analyze a VICES functional check result
I1030	Analyze a VICES test/trial/inspection result or report
M0700	Analyze/evaluate/interpret a Global Positioning System (GPS) functional check result
M0710	Analyze/evaluate/interpret a GPS system test/trial/inspection result or report
M1030	Direct a corrective maintenance/repair activity on a GPS system
M1070	Direct a GPS functional check
M1080	Direct a GPS system PM activity (including a test/trial/inspection)
M0870	Analyze/evaluate/interpret a NAV Radar system functional check result
M0880	Analyze/evaluate/interpret a NAV Radar system test/trial/inspection result or report
M1150	Direct a corrective maintenance/repair activity on a NAV Radar system
M1160	Direct a NAV Radar System functional check

M1170	Direct a NAV Radar system PM activity (including a test/trial/inspection)
J0390	Analyze/evaluate/interpret an HF/MF/LF/VLF Transmit System functional check result
J0400	Analyze/evaluate/interpret an HF/MF/LF/VLF Transmit System test/trial/inspection result or report
J1140	Direct a corrective maintenance/repair activity on an HF/MF/LF/VLF Transmit System
J1250	Direct an HF/MF/LF/VLF Transmit System functional check
J1260	Direct an HF/MF/LF/VLF Transmit System PM activity (including a test/trial/inspection)
J0150	Analyze/evaluate/interpret a VHF System test/trial/inspection result or report
J0160	Analyze/evaluate/interpret a VHF System test/trial/inspection results/report
J1100	Direct a corrective maintenance/repair activity on a VHF System
J1170	Direct a VHF System functional check
J1180	Direct a VHF System PM activity (including a test/trial/inspection)
J0330	Analyze/evaluate/interpret an HF/MF/LF/VLF Receive System functional check result
J0340	Analyze/evaluate/interpret an HF/MF/LF/VLF Receive System test/trial/inspection result or report
J1130	Direct a corrective maintenance/repair activity on an HF/MF/LF/VLF Receive System
J1230	Direct an HF/MF/LF/VLF Receive System functional check
J1240	Direct an HF/MF/LF/VLF Receive System PM activity (including a test/trial/inspection)
K0880	Analyze/evaluate/interpret a Radio Antenna System and Structure system test/trial/inspection result or report
K0920	Direct a corrective maintenance/repair activity on a Radio Antenna System and Structure
K0960	Direct a corrective maintenance/repair activity on a Radio Antenna System and Structure
K1000	Direct a Radio Antenna System and Structure functional check
K1010	Direct a Radio Antenna System and Structure PM activity (including a test/trial/inspection)
O0630	Analyze/evaluate/interpret a JOTS/JMICS/GCCS/EDO LINK System functional check result
O0640	Analyze/evaluate/interpret a JOTS/JMICS/GCCS/EDO LINK System test/trial/inspection result or report
O0870	Direct a corrective maintenance/repair activity on a JOTS/JMICS/GCCS/EDO LINK System
O0910	Direct a JOTS/JMICS/GCCS/EDO LINK System functional check
O0920	Direct a JOTS/JMICS/GCCS/EDO LINK System PM activity (including a test/trial/inspection)
G1450	Direct a 2046 sonar functional check
G1460	Direct a 2046 sonar corrective maintenance
G1470	Direct a 2046 sonar preventive maintenance
G1480	Analyze a 2046 sonar functional check result
G1490	Analyze a 2046 sonar test/trial/inspection result or report
N1220	Direct a time server functional check result
N1230	Direct a time server corrective maintenance

N1240	Direct a time server preventive maintenance
N1250	Analyze a time server functional check result
N1260	Analyze a time server test/trial/inspection result or report
J0110	Analyze/evaluate/interpret a UHF System functional check result
J0120	Analyze/evaluate/interpret a UHF System test/trial/inspection result or report
J1090	Direct a corrective maintenance/repair activity on a UHF System
J1150	Direct a UHF System functional check
J1160	Direct a UHF System PM activity (including a test/trial/inspection)
K0910	Analyze/evaluate/interpret a Satellite Communications System functional check result
K0950	Analyze/evaluate/interpret a Satellite Communications System test/trial/inspection result or report
K1030	Direct a Satellite Communications System functional check
K1040	Direct a Satellite Communications System PM activity (including a test/trial/inspection)
M1220	Direct a NDDS functional check
M1230	Direct a NDDS corrective maintenance
M1240	Direct a NDDS preventive maintenance
M1250	Analyze a NDDS functional check result
M1260	Analyze a NDDS test/trial/inspection result or report
G1390	Direct a BQQ/10 functional check
G1400	Direct a BQQ/10 corrective maintenance
G1410	Direct a BQQ/10 preventive maintenance
G1420	Analyze a BQQ/10 functional check result
G1430	Analyze a BQQ/10 test/trial/inspection result or report
H1140	Direct a 2073 EUWT functional check
H1150	Direct a 2073 EUWT corrective maintenance
H1160	Direct a 2073 EUWT preventive maintenance
H1170	Analyze a 2073 EUWT functional check result
H1180	Analyze a 2073 EUWT test/trial/inspection result or report
H1200	Direct a 2071 Augmenter functional check
H1210	Direct a 2071 Augmenter corrective maintenance
H1220	Direct a 2071 Augmenter preventive maintenance
H1230	Analyze a 2071 Augmenter functional check result
H1240	Analyze a 2071 Augmenter test/trial/inspection result or report
H1260	Direct a 189 Cavitation Indicator functional check
H1270	Direct a 189 Cavitation Indicator corrective maintenance

H1280	Direct a 189 Cavitation Indicator preventive maintenance
H1290	Analyze a 189 Cavitation Indicator functional check result
H1300	Analyze a 189 Cavitation Indicator test/trial/inspection result or report
H1320	Direct a Speed Dependent Tonal system functional check
H1330	Direct a Speed Dependent Tonal system corrective maintenance
H1340	Direct a Speed Dependent Tonal system preventive maintenance
H1350	Analyze a Speed Dependent Tonal system functional check result
H1360	Analyze a Speed Dependent Tonal system test/trial/inspection result or report
H1070	Direct a SSE System functional check
H1100	Analyze/evaluate/interpret a SSE System functional check result
H1110	Analyze/evaluate/interpret a SSE System test/trial/inspection result or report
H1380	Direct an SSE corrective maintenance
H1390	Direct an SSE preventive maintenance
L1060	Direct cryptographic equipment functional check
L1070	Direct cryptographic equipment corrective maintenance
L1080	Direct cryptographic equipment preventive maintenance
L1090	Analyze cryptographic equipment functional check result
L1100	Analyze cryptographic equipment test/trial/inspection result or report
E1290	Direct a SeaSearch II functional check
E1300	Direct a SeaSearch II corrective maintenance
E1310	Direct a SeaSearch II preventive maintenance
E1320	Analyze a SeaSearch II functional check result
E1330	Analyze a SeaSearch II test/trial/inspection result or report
H0010	Analyse a Submarine WSDB test/trial/inspection result or report
K0930	Direct a corrective maintenance/repair activity on a Satellite Communications System
K1080	Direct an ISDS/Sub IP functional check
K1090	Direct an ISDS/Sub IP corrective maintenance
K1100	Direct an ISDS/Sub IP preventive maintenance
K1110	Analyze an ISDS/Sub IP functional check result
K1120	Analyze an ISDS/Sub IP test/trial/inspection result or report
I1050	Direct a VCS 446 functional check
I1060	Direct a VCS 446 corrective maintenance
I1070	Direct a VCS 446 preventive maintenance

I1080	Analyze a VCS 446 functional check result
I1090	Analyze a VCS 446 test/trial/inspection result or report
I1110	Direct a Sound Powered Network functional check
I1120	Direct a Sound Powered Network corrective maintenance
I1130	Direct a Sound Powered Network preventive maintenance
I1140	Analyze a Sound Powered Network functional check result
I1150	Analyze a Sound Powered Network test/trial/inspection result or report

b. Recommend this EO for DL development;

c. TP2 – Trainees to read NAVORD 3250-7 (RCN Material OPDEF Process) prior to instruction; and

d. Guest lecturers - Representatives from FTA and/or ST(S).

10. Teaching Points

	Description	Instructional Method	Time	Reference

TP1	Introduction to unit-level maintenance
	- TRP
	- (S1330) Supporting escape and rescue certificate
	- (S1320) Submarine Search and Rescue (SUBSAR) requirements
	- Outside organizations
	  o Purpose
	  o Contacts
	  o Organizational charts
	- Objective Quality Evidence (OQE) by FTA
	- Victoria-Class Submarine Technical Instruction (VSTI)
	- Submarine Temporary Instruction (SMTI)
	- First-Level Area Diagram (FLAD)
	
	CS Maintenance Procedures
	1. Direct equipment functional check
	2. Analyze equipment functional check result or report
	3. Analyze equipment test/trial/inspection result or report
		Guest lecture
		
		Interactive lecture
			3 x 50 mins	A6, A4, A9

	4. Analyze a CS suite level test/trial/inspection result or report 5. Direct equipment CM/repair activity 6. Direct equipment PM activity			
TP2	Coordinate CM - (A1240) Direct the repair of CS equipment faults - (A1270) Provide options/recommendations on CS equipment CM to Command - (S1330) Monitor CSED support IAW Escape and Rescue Certificate - (A0190) Draft an OPDEF - (A0990) Coordinate departmental CM activities - (A1290) Evaluate CS faults - FTA defect vetting - Fleet Maintenance Facilities Service Delivery Process (Notification process) - Direct a CM activity for CS equipment (see para 9a)	Guest lecture Case study Guided discussion Self-study	7 x 50 mins	A6, A10, A87, A92

TP3	(A1250) Coordinate system trials / (A1300) Coordinate the conduct of weapons system engineering test forms and trials
	- Trial types
	  ○ Test forms
	  ○ Set-to-work (STW)
	  ○ Harbour Acceptance Trial (HAT)
	  ○ Sea Acceptance Trial (SAT)
	  ○ Safe-to-dive Trial
	- Requirements
	- Process by FTA Naval Engineering & Maintenance
	- Liaising with FTA
	- Liaising with unit combat officer
	- (A1110) Analyze a ship-level test trial/inspection result/report
	- (A0860) Analyze CS suite-level test trial/inspection result/report
	  ○ Communication Suite
	  ○ EW suite
	  ○ Navigation Suite
	  ○ Sonar Suite
		Guest lecture
		
		Interactive lecture
			3 x 50 mins	A6

	○ Fire Control Suite ○ CS Integration Suite - Analyze a CS equipment test trial/inspection result/report (see para 9a)			
TP4	Coordinate PM - Maintenance cycle - Work period timelines - (A1230) Direct the conduct of CS PM - (A0830) Track and report work period progress - (A1000) Coordinate departmental PM activities - (A0120) Draft a PM schedule - (A0040) Resolve an outside agency assistance problem - Direct a PM activity for CS equipment (see para 9a)	Guest lecture Interactive lecture	2 x 50 mins	A6, A7, A53, A54, A55, A92

Enabling Check - EO 003.02 - Coordinate unit-level maintenance activities

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom; and

2. Laptop / workstation with DRMIS.

Arrangements and Procedures

3. EC 003.02 will be a take-home written assessment based on content from EO 003.02.

4. Pass mark will be 70%.

5. Time required: 1 x 50 min period (review / debrief)

6. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.02 will be a take-home written assessment based on content from EO 003.02.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

3. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

4. Read each question carefully before answering.

5. References will be provided.

6. Pass mark is 70%.

7. You will have 1 x 50 min period (review / debrief).

8. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Take-home assignment to include a minimum of 20 short-answer questions, where the trainee must answer and provide the supporting reference/policy.

PART IV – ENABLING CHECK LIST

Assessment Items	Percentage
TP1 - Introduction to unit-level maintenance	10
TP2 - CM	50
TP3 - System Trials	10
TP4 - PM	30

The trainee must achieve 70% to pass this EC.

PO 003 Coordinate Divisional Technical Readiness

EO 003.03 Manage CSED Pre-Sail Checks

1. Performance. Manage CSED pre-sail checks.

2. Conditions

   a. Given:

      (1) Laptop / Workstation with DRMIS; and

      (2) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall (A0450) manage CSED pre-sail checks, IAW references, by:

   a. Directing departmental pre-sail checks;

   b. (A0210) Evaluating pre-sail check results; and

   c. Briefing the Material State.

4. Time Required

   a. EO - 2 x 50 min period; and

   b. EC - 1 x 50 min period (review / debrief).

5. Instructional Methods. Interactive lecture.

6. Training Aids

   a. Audio/visual aids; and

   b. Sample Material State Brief.

7. Test Details. EO 003.03 will be assessed in EC 003.03 as a take home assignment, and in PC 003.

8. References. See para 10.

9. Remarks. Recommend this EO for DL development.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Directing departmental pre-sail checks - (A1170) Direct CSE communication pre-sail checks - (A1200) Direct CSE fire control pre-sail checks - (A1160) Direct CSE sonar pre-sail checks - Reports - Timelines	Interactive lecture	0.5 x 50 mins	A53, A54, A55, A92
TP2	(A0210) Evaluate/interpret pre-sail check results - Prioritization - Implications	Interactive lecture	0.5 x 50 mins	A53, A54, A55

TP3	Material State brief
	- Contents
	  ○ OPDEFs
	  ○ Non-standard line-ups or operating procedure
	  ○ Documentation or stores deficiencies
	  ○ Weapon readiness state
	  ○ Ammunition
	  ○ Outstanding trials
	  ○ Personnel and qualifications
	- Timing and duration
	- Overview
		Interactive lecture	1 x 50 mins	A87, A92, A94

Enabling Check - EO 003.03 - Manage CSED pre-sail checks

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 003.03 will be a take-home practical assessment based on content from EO 003.03.

3. Pass mark will be 6 of 7 SAT.

4. Time required: 1 x 50 min period (review / de-brief).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.03 will be a take-home practical assessment based on content from EO 003.03.

2. During this practical assessment, you will be required to Manage CSED pre-sail checks. You must pass this assessment to pass EO 003.03.

3. References will be provided.

4. Pass mark is 6 of 7 SAT.

5. You will have 1 x 50 min period (review / de-brief).

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Scenario based assessment, whereby trainees must complete a Material State brief by interpreting detailed pre-sail check lists and other pertinent information such as OPDEFs, Commanding Officer Temporary Memorandums (COTMs), maintenance completion rates, etc.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Complete Material State Brief IAW VCSSO to include:			
Existing OPDEFs			

Performance Factors	SAT	UNSAT	Comments
Non-standard line-ups or operating procedure			
Documentation or stores deficiencies			
Weapon readiness state			
Ammunition			
Outstanding trials			
Personnel and qualifications			

The trainee must achieve 6 of 7 SATs to pass this EC.

PO 003 Coordinate Divisional Technical Readiness

EO 003.04 Support Towed Array Operations.

1. Performance. Support towed array operations.

2. Conditions

   a. Given:

      (1) Laptop / workstation with DRMIS; and

      (2) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall (S1610) support towed array operations, IAW references, by:

   a. (S1340) Monitoring CS department for towed array operations; and

   b. Supporting towed array pre/post-deployment preparations.

4. Time Required

   a. EO - 2 x 50 min period; and

   b. EC - 1 x 50 min period (review / debrief).

5. Instructional Method. Interactive lecture.

6. Training Aids.

   a. Audio/visual aids; and

   b. Deployment and recovery video.

7. Test Details. EO 003.04 will be assessed in EC 003.04 as a self-study assignment, and in PC 003.

8. References. See para 10.

9. Remarks. Recommend this EO for DL development.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	(S1340) Monitor CS department for towed array operations	Interactive lecture	1 x 50 mins	A58

	- Towed array deployment capabilities and limitations - Towed array release - Deployment and recovery overview - Towed array deployment equipment			

TP2	Support towed array pre/post-deployment preparations
	- Trim considerations
	- Diver support
	- Submarine Towed Array Handling Team (STAHT)
	- Functional checks
	  ○ Umbilical
	  ○ Towed array
		Interactive lecture	1 x 50 mins	A58, A59, A60, A61, A62, A93

Enabling Check - EO 003.04 - Support Towed Array Operations

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 003.04 will be a take-home written assessment based on content from EO 003.04.

3. Pass mark will be 70%.

4. Time required: 1 x 50 min period (review / debrief).

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.04 will be a written assessment based on content from EO 003.04.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

3. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

4. Read each question carefully before answering.

5. References will be provided.

6. Pass mark is 70%.

7. You will have 1 x 50 min period (review / debrief).

8. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Take home assignment to include a minimum of ten short-answer questions. References must be provided with answers.

PART IV – ENABLING CHECK LIST

Assessment Items	Percentage
TP1 – Monitor CSED for towed array operations	50
TP2 – Support towed array pre/post-deployment preparations	50

The trainee must achieve 70% to pass this EC.

PO 003 Coordinate Divisional Technical Readiness

EO 003.05 Contribute to the Project Management Process

1. Performance. Contribute to the Project Management Process.

2. Conditions

   a. Given:

      (1) References; and

      (2) Project Management forms.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall contribute to the project management process, IAW references, by:

   a. Reviewing project management forms;

   b. Providing subject matter expertise to the project management PM process; and

   c. Managing project requirements.

4. Time Required

   a. EO - 10 x 50 min periods; and

   b. EC - 1 x 50 min period.

5. Instructional Methods. Interactive Discussion.

6. Training Aids. Sample Project Documentation (SOCD, SOW, etc.)

7. Test Details. EO 003.05 will be assessed in EC 003.05 and/or in the CSPS course offerings (see para 9).

8. References. See para 10.

9. Remarks. Recommend this EO for DL development. Risk/Project Management will soon be offered via DL through the Canada School of Public Service (CSPS), therefore training timings will be adjusted accordingly IAW the CSPS offerings, as follows. Not all modules will apply to the CSEC/O therefore staff must review and select applicable modules for training IAW para 10 Teaching Points:

Offering	Session	Delivery	Duration
Introduction to Risk Management	00047484 (E) 00077485 (F)	CSPS	2:00
Risk Response, Monitor and Control	00073838 00073840	CSPS	2:00
Introduction to Project Management	00077011 00077012	CSPS	1:00
Defining and sequencing project Activities	00073625 00073626	CSPS	2:00
Integrated Project Execution, Monitoring and Control	00073601 00073602	CSPS	1:30
Managing Project Human Resources	00073809 00073811	CSPS	2:00
Developing and Controlling Project Schedule	00073735 00073737	CSPS	2:00
Create Work Breakdown Structure	00073616 00073619	CSPS	1:30
Quality Assurance and Quality Control	00073775 00073776	CSPS	2:00

10. Teaching Points

	Description	Instructional Method	Time	References
TP1	PM Forms / Terminology - Statement of Work (SOW) - Statement of Capability Deficiency (SOCD) - Statement of Requirements (SOR) - Technical Statement of Requirements (TSOR) - Statement of Work (SOW) - Request of Proposals (RFP)	Interactive Lecture	1 x 50 mins	C6, A90

TP2	PM Process
	- Initiating
	- Planning
	  o Options analysis
	- Executing
	  o Trials
	- Monitoring / Controlling
	- Closing
		Interactive Lecture	2 x 50 mins	C6, A90

TP3	Project Stakeholders, including the roles and importance of each - Project Manager	Interactive Lecture	2 x 50 mins	C6, A90

	- Customer - Performing Organization - Team members - Sponsor			
TP4	Project Management Requirements - Integration Management - Scope Management - Time Management - Cost Management - Quality Management - Human Resource Management - Communications Management - Risk Management - Procurement Management	Interactive Lecture	1 x 50 mins	C6, A90

TP5	Project Scope Management
	- Project Charter
	  o Contents
	  o Use
	  o Importance
	- Project Plan
	  o Contents
	  o Use
	  o Importance
		Interactive Lecture	0.5 x 50 mins	C6, A90

TP6	Project Time Management; - Activity Sequencing - Activity Duration Estimating - Schedule baselines - Change control - Briefing Command on Project progress	Interactive Lecture	1 x 50 mins	C6, A90
TP7	Project Quality Management - Quality Assurance/Control-purpose, definitions, differences - Standards	Interactive Lecture	0.5 x 50 mins	C6, A90

	- Control charts			
TP8	Project Risk Management - Risk Identification - Risk Analysis - Response Planning - Risk Control	Interactive Lecture	2 x 50 mins	C6, A90

Enabling Check - EO 003.05 - Contribute to the Project Management Process

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 003.05 will be a scenario-based written assessment based on content from EO 003.05.

3. Pass mark will be 70%.

4. Time required: 1 x 50 min period.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.05 will be a scenario-based written assessment based on content from EO 003.05.

2. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

3. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

4. Read each question carefully before answering.

5. References will be provided.

6. Pass mark is 70%.

7. You will have 1 x 50 min period.

8. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. EC to include sample PM forms whereby the trainee must identify the form, and interpret the information provided. EC is to be a minimum of ten scenario-based, short-answer questions.

PART IV – ENABLING CHECK LIST

Assessment Item	Percentage
TP1 - PM Forms / Terminology	20
TP2 - PM Process	20
TP3 - Project Stakeholders	5
TP4 - Project Management Requirements	5
TP5 - Project Scope Management	5
TP6 - Project Time Management	20
TP7 - Project Quality Management	5
TP8 - Project Risk Management	20

The trainee must achieve 70% to pass this EC.

PERFORMANCE CHECK - PO 003 - Coordinate Divisional Technical Readiness

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom; and

2. Laptop / workstation, with Message Drafter and/or DRMIS.

Arrangements and Procedures

3. PC 003 will be a practical assessment based on content from PO 003.

4. Pass mark will be 7 of 10 SAT, including SAT on all critical performance factors.

5. Time required: 1 x 50 min period.

6. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a practical assessment based on content from PO 003.

2. During this practical assessment, you will be required to coordinate divisional technical readiness. You must pass this assessment to pass PO 003.

3. References will be provided.

4. Pass mark is 7 of 10 SAT, including SAT on all critical performance factors.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Practical, scenario-based assessment. The trainee will be provided a narrative containing all information required to fill out an OPDEF on faulted equipment. Trainee will then complete the OPDEF IAW the OPDEF standard.

PART IV – PERFORMANCE CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Addressees			
OPDEF category			
OPDEF definition			

Performance Factors	SAT	UNSAT	Comments
Situation			
Action taken			
Ship's staff intentions			
Operational impact			
Stores details			
OPDEF fully populated*			
Correct Message format*			

* Indicates a critical performance factor. The trainee must achieve 7 of 10 SAT including SAT on all critical performance factors to pass this PC.

PO 004 Support Weapon Certification Programme

EO 004.01 Support WCP Administration

1. Performance. Support WCP Administration.

2. Conditions

   a. Given: References.

   b. Denied:

      (1) Assistance; and

      (2) Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall (A1320) support WCP administration, IAW references, by:

   a. Applying WCP Policy;

   b. Overseeing the WCP;

   c. Managing Weapon System Training;

   d. Managing Technical Weapon System Certification;

   e. Validating the Weapon System Capability; and

   f. Managing the Data Collection Plan.

4. Time Required

   a. EO - 6 x 50 min periods;

   b. EC - 1 x 50 min period; and

   c. PC - 7 x 50 min periods (1 period / trainee (6) for the practical assessment + 1 period for written assessment).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion;

   c. Self-study; and

   d. Case study.

6. Training Aids. Nil.

7. Test Details. EO 004.01 will be assessed in EC 004.01 and in PC 004.

8. References. See para 10.

9. Remarks

a. TPs are a direct review of CFCD 122, by chapter;

b. Self-study pre-reading of CFCD 122 and NAVORDs prior to beginning this EO; and

c. Recommend this EO for DL development.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Weapon Certification Programme Policy	Interactive lecture / Guided Discussion	0.5 x 50 mins	A4, A81, A86, A88
TP2	WCP	Interactive lecture / Guided Discussion / Case Study	1.5 x 50 mins	A81, A86, A88
TP3	Weapon System Training	Interactive lecture / Guided Discussion	1.5 x 50 mins	A81, A86, A88
TP4	Technical Weapon System Certification	Interactive lecture / Guided Discussion	1 x 50 mins	A81
TP5	Weapon System Capability Validation	Interactive lecture / Guided Discussion	1 x 50 mins	A81, A86, A88
TP6	Data Collection Plan	Interactive lecture / Guided Discussion	0.5 x 50 mins	A81

Enabling check - EO 004.01 - Support WCP Administration

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 004.01 will be a written assessment based on content from EO 004.01.

3. Pass mark will be 70%.

4. Time required: 1 x 50 min period.

5. References will be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.01 will be a written assessment based on content from EO 004.01.

2. Ensure your desk is clear of material except for the test booklet and / or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).

6. Read each question carefully before answering.

7. If you have any questions during this assessment, raise your hand.

8. References will be provided.

9. Pass mark is 70%.

10. You will have 1 x 50 min period to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Written assessment consisting of a minimum ten short-answer questions. References must be provided with answers.

PART IV – ENABLING CHECK LIST

Assessment Items	Percentage
TP1 – Weapon certification programme policy	18
TP2 – Weapon certification programme	18
TP3 – Weapon system training	18
TP4 – Technical weapon system certification	18
TP5 – Weapon system capability validation	18
TP6 – Data collection plan	10

The trainee must achieve 70% to pass this EC.

PO 004 Support WCP

EO 004.02 Support WHDS Training.

1. Performance. Support WHDS training.

2. Conditions

   a. Given:

      (1) WHDST;

      (2) TWS Trainer;

      (3) Equipment, IAW refs;

      (4) References; and

      (5) Supervision.

   b. Denied: Assistance.

   c. Environmental:

      (1) Classroom;

      (2) WHDST; and

      (3) TWS Trainer.

3. Standard. The trainee shall (S1620) support WHDS training, IAW references, by:

   a. Conducting pre-embarkation checks / preparation for embarkation;

   b. (F0740) Supervising the embarking/disembarking of a torpedo;

   c. (S1310) Conducting a receipt / transfer inspection of Mk. 48 torpedo; and

   d. Supervise loading / unloading.

4. Time Required

   a. EO - 27 x 50 min periods; and

   b. EC - 9 x 50 min periods (1.5 periods / trainee).

5. Instructional Methods

   a. Interactive lecture; and

   b. Demonstration - performance.

6. Training Aids

a. Audio/visual aids;

b. Annexes A & B of CFCD 122; and

c. IT resources.

7. Test Details. EO 004.02 will be assessed in EC 004.02 and PC 004.

8. References. See para 10.

9. Remarks

a. Staff shall demo an evolution from acceptance of weapon to delivery and re-delivery, including pre-loading, loading, and casualty procedures; and

b. TP3 - During individual receipt inspections, trainees not involved will review references, assist (as required), and/or conduct remaining elements of TP.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Pre-embarkation checks / Preparations for embarkation - SUBWOPS safety summary - (F0020) Embarkation / disembarkation team - Unpacking/packing torpedo from transport container - (B0270) Torpedo log / record keeping - OFM - Unpacking/packing torpedo-mounted dispenser from transport container	Interactive lecture / Demonstration / Performance	2 x 50 mins 4 x 50 mins (Full demo)	A8, A45, A53 (PG 121)
TP2	(F0740) Supervising the embarking / disembarking of a torpedo - Responsibilities of CSEC/O - Embarkation trolley fitting / removal - Nose / tail guidelines fitting / removal - Casualty procedures for embarking	Interactive lecture / Demonstration	2 x 50 mins	A53 (PG 124)

TP3	(S1310) Torpedo receipt / transfer inspection - Torpedo receipt inspection - (S1640) Torpedo-mounted dispenser (TMD) inspection - TMD installation - (F0150) Control Cable fitting/ removal - Guidance wire splicing - Guidance wire connection / disconnection - (F0760) Band / de-band a torpedo	Interactive lecture / Demonstration / Performance	15 x 50 mins (2.5 x inspection per trainee)	A53 (OP 120)
TP4	Supervise loading / unloading - Transfer torpedo to the Weapon Stowage Compartment (WSC) - Post-load diagnostic test of Mk. 48 - Casualty Procedures for loading - Casualty Procedures for hangfire/ misfire	Interactive lecture / Demonstration	4 x 50 mins	A53 (OP 129), A54

Enabling Check - EO 004.02 - Support WHDS Training

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. WHDST.

Arrangements and Procedures

2. EC 004.02 will be a two-part practical assessment based on content from EO 004.02.

3. Pass mark will be all SAT.

4. Time required: 9 x 50 min periods (1.5 periods / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.02 will be a two part practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to support WHDS training. You must pass this assessment to pass EO 004.02.

3. References will be provided.

4. Pass mark will be all SAT.

5. You will have 1.5 x 50 min periods to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical assessment will consist of two parts:

   a. EC Part A. Trainee will perform a receipt inspection of a weapon IAW SUBWOPS PG124 step 1 (OP 120 Part A – I) and jetty party actions (PG124 steps 8 – 11), while respecting SUBWOPS safety summary.

   b. Note: Two personnel will be required to assist jetty team.

   c. EC Part B. Trainee will participate in a job-based tabletop activity consisting of realistic scenarios encountered on the job. This will assess the ability of the trainees to supervise the pre-embarkation checks, the embarking, disembarking, loading, and unloading of a Mk. 48 torpedo. Trainees will be required to take action on discovery of safety infraction and/or Casualty Procedure (CP) IAW SUBWOPS.

PART IV – ENABLING CHECK LIST

1. EC Part A - EC List can be found SUBWOPS Vol. 1 PG124 Steps A - I. The trainee must complete all steps of a receipt inspection IAW SUBWOPS to achieve SAT on the Receipt Inspection performance factor in the table below.

2. EC Part B – IAW EC table:

Performance Factors	SAT	UNSAT	Comments
PART A – Receipt Inspection			
Receipt inspection IAW SUBWOPS			
PART B – Scenario-based Tabletop			
Detail team requirements			
Casualty Procedures (embarking/disembarking)			
Casualty Procedures (loading / unloading)			
Safety IAW SUBWOPS			

PO 004 Support Weapon Certification Programme

EO 004.03 Support Weapons Employment for WCP

1. Performance. Support weapons employment for WCP.

2. Conditions

   a. Given:

      (1) Victoria Class Submarine Command Team Trainer (VSCTT); and

      (2) References.

   b. Denied:

      (1) Supervision; and

      (2) Assistance.

   c. Environmental:

      (1) VSCTT; and

      (2) Secure discussion area.

3. Standard. The trainee shall (S1630) support weapons employment for WCP, IAW references, by:

   a. Supervising Weapon Control; and

   b. Overseeing a torpedo launch.

4. Time Required

   a. EO - 20 x 50 min periods; and

   b. EC - 6 x 50 min periods (1 period / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Demonstration - performance; and

   c. Self-study.

6. Training Aids. Sample Mk. 48 Build Message.

7. Test Details. EO 004.03 will be assessed in EC 004.03 and PC 004.

8. References. See para 10.

9. Remarks

a. This EO contains classified material;

b. TP 2 - CSEC/O will not perform the task of firing a torpedo, however they are expected to be the SME for all weapons related activities;

c. Self-study to be conducted in secure discussion area due to publication classification. The trainees are expected to familiarize themselves with all relevant reference material listed in para 10; and

d. The Threshold Knowledge Test contains Secret information and must be stored according to its classification.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Weapon / Weapon System Training (WST) Overview - Capabilities and limitations - Characteristics - Weapon build message - WST threshold knowledge test orientation	Interactive lecture Guided discussion Self-study	6 x 50 mins	A46, A47
TP2	SFCS Weapon Control Overview - Firing drills - Weapons readiness states - Weapon discrete signals - Pre/post-launch settings - Technical weapon employment	Demonstration - performance	7 x 50 mins	A49, A54, A80
TP3	Oversee a torpedo launch / Launch a torpedo - Exercise mode / onboard training (OBT) - (F0040) Primary mode - Secondary mode - (F0030) Casualty procedures / Emergency Mode o (F0880) Torpedo Launching System misfire routine	Demonstration - performance Self-study	7 x 50 mins	A49, A53, A54, A80

Enabling Check - EO 004.03 - Support Weapons Employment for WCP

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Fred Sherwood Trainer.

Arrangements and Procedures

2. EC 004.03 will be a practical assessment based on content from EO 004.03.

3. Pass mark will be 14 of 17 SAT, including SAT on all critical performance factors.

4. Time required: 6 x 50 min periods (1 period / trainee).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.03 will be a practical assessment based on content from EO 004.03.

2. During this practical assessment, you will be required to support weapons employment for WCP. You must pass this assessment to pass EO 004.03.

3. References will not be provided.

4. Pass mark is 14 of 17 SAT, including SAT on all critical performance factors.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Trainees shall conduct a deliberate attack up to and including the walking back from the engagement, providing an explanation of each step as they carry them out.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Inventory weapon (Tube order matrix)			
Bring weapon system to patrol state			
Bring weapon system to action state			

Performance Factors	SAT	UNSAT	Comments
Choose threat			
Allocate tube			
Select tactic			
Warm-up weapon			
Apply pre-sets			
Exercise weapon			
Select tube			
Safe/Fire key to Fire* **- Explain signals required** **- Explain hangfire and misfire and their follow-up actions**			
Fire weapon* **- Explain post-firing sequence**			
Safe/ Fire key to Safe			
Enable weapon			
Change a multi-state pre-set			
Shut-down weapon			
Walk back from engagement			

*Indicates a critical performance factor. Trainee must achieve 14 of 17 SAT, including SAT on all critical performance factors in order to meet the standard of this EC.

PERFORMANCE CHECK - PO 004 - Support Weapon Certification Programme

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Secure discussion area; and

2. WHDS Trainer.

Arrangements and Procedures

1. PC 004 will be a two-part practical assessment based on content from PO 004.

2. Part 1: Pass mark will be 70%. Part 2: Pass mark will be all SAT.

3. Time required: 7 x 50 min periods (1 period for Threshold Knowledge Test and 1 period / trainee practical).

4. References will be required.

5. Threshold Knowledge Test contains secret information and must be stored according to its classification.

PART II – PC PART 1- WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 Part 1 will be a Threshold Knowledge Test, based on content from PO 004, and as administered on the job.

2. During this practical assessment, you will be required to support the weapon certification programme. You must pass this assessment to pass PO 004.

3. References will not be provided.

4. Pass mark is 70%.

5. You will have 1 x 50 min period to complete the Threshold Knowledge Test.

6. Do you have any questions before you start?

PART II – PC PART 2 - PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 Part 2 will be a practical assessment based on content from PO 004.

2. During this practical assessment, you will be required to support the WCP. You must pass this assessment to pass PO 004.

3. References will be provided.

4. Pass mark is all SAT.

5. You will have 1 x 50 min period to complete the practical assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. PC Part 1. Written Threshold Knowledge Test - IAW CFCD 122. This test is classified.

2. PC Part 2 – Practical assessment whereby the trainee must perform a receipt inspection of a weapon IAW SUBWOPS OP 120 Part A – G including correct data collection sheets while respecting SUBWOPS safety summary. (CFCD 122 Annex A & B). The trainee must complete all steps of the receipt inspection IAW SUBWOPS and complete data collection sheets.

PART IV – PERFORMANCE CHECK LIST

Part 2 – Practical - PC List IAW SUBWOPS Vol. 1 OP 120 Steps A - G.

PO 005 Conduct duties of CS SME in HQ1

EO 005.01 Sustain Combat Systems in Emergency Situations

1. Performance. Sustain combat systems in emergency situations.

2. Conditions

   a. Given:

      (1) References; and

      (2) Assistance.

   b. Denied: Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall sustain combat systems in emergency situations, IAW references, by:

   a. (A1080) Diagnosing CS suite-level faults;

   b. (B0860) Directing CS ERT; and

   c. (A0620) Providing recommendations to Command on CS equipment availability and reconfiguration options.

4. Time Required

   a. EO - 35 x 50 min periods;

   b. EC - 6 x 50 min periods (1 period / trainee); and

   c. PC - 6 x 50 min periods (1 period / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion;

   c. Self-study; and

   d. Role play.

6. Training Aids. Electrical audit handouts.

7. Test Details. EO 005.01 will be assessed in EC 005.01 and PC 005.

8. References. See para 10.

9. Remarks

a. Recommended support manning for practical-based scenarios in TP3 are detailed in Chapter 1, Special Instructions.

b. Trainees are to familiarize themselves with reference material via self-study.

10. Teaching Points

	Description	Instructional Method	Time	Reference

TP1	(A1080) Diagnose suite-level faults
	- Anatomy of a Damage Control (DC) incident
	- HQ1 publications
	- Power distribution on Victoria Class
	  o Fuse panel naming convention
	  o Load shedding
	- Electrical audit
	- Ship's system considerations (i.e. Hydraulics, air, etc.)
	- (I0230) Perform a Public Address System functional check
		Interactive lecture
		
		Role play
		
		Guided discussion
		
		Self-study
			10 x 50 mins	A72, A73, A74, A75, A76, A77, A78, A79, A91, A92, A93

TP2	(B0860) Direct CS ERT - DC organization - Employing ERT - Briefing ERT	Interactive lecture Role play Guided discussion	4 x 50 mins	A72, A73, A74, A75, A76, A77, A78, A79, A92
TP3	(A0620) Provide recommendations to Command on CS equipment availability and reconfiguration options - DC Scenarios - HQ1 communication / briefing incident controller (IC)	Role play Guided discussion	21 x 50 mins	A72, A73, A74, A75, A76, A77, A78, A79, A91, A92

Enabling Check - EO 005.01 - Sustain Combat Systems in Emergency Situations

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 005.01 will be a practical assessment based on content from EO 005.01.

3. Pass mark will be all SAT.

4. Time required: 6 x 50 min periods (1 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.01 will be a practical assessment based on content from EO 005.01.

2. During this practical assessment, you will be required to sustain Combat Systems in emergency situations. You must pass this assessment to pass EO 005.01.

3. References will be provided.

4. Pass mark is all SAT.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical EC will consist of the trainee acting as a CSE SME in HQ1 during an emergency scenario. Instructor will fill all other roles as required during the scenario. Repairs must be complete and logical IAW scenario.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Close-up to include ERT			
Brief Incident Controller on losses to CS equipment			

Performance Factors	SAT	UNSAT	Comments
Request / receive Command priority			
Provide recommendations and plan of action			
Implement plan of action (i.e. Change-Over Switches (COS))			
Revert to SOP line-up			
Direct an electrical audit			
Fault-find IAW HQ1 publications to determine the faults			
Task ERT to complete repairs			

The trainee must achieve all SAT to pass this EC.

PERFORMANCE CHECK - PO 005 - Conduct Duties of CS SME in HQ1

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 005 will be a practical assessment based on content from PO 005.

3. Pass mark will be all SAT.

4. Time required: 6 x 50 min periods (1 period / trainee).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 005 will be a practical assessment based on content from PO 005.

2. During this practical assessment, you will be required to conduct the duties of CS SME in HQ1. You must pass this assessment to pass PO 005.

3. References will be provided.

4. Pass mark is all SAT.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Trainees must act as CS SME in HQ1 during an emergency scenario. Instructor will fill all other roles as required during scenario. Repairs must be complete and logical IAW scenario. Scenario must be different than EC 005.01.

PART IV – PERFORMANCE CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Close-up to include ERT			
Brief Incident Controller on losses to CS equipment			

Performance Factors	SAT	UNSAT	Comments
Request / achieve Command priority			
Provide recommendations and plan of action			
Implement plan of action (i.e. COS)			
Revert to SOP line-up			
Direct an electrical audit			
Fault-find IAW HQ1 publications to determine the faults			
Task ERT to complete repairs			

The trainee must achieve all SAT to pass this PC.

PO 006 Advise Command on the CS Suite

EO 006.01 Advise Command on the Communication Suite

1. Performance. Advise Command on the communication suite.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;
      (2) Supervision;
      (3) Engineering drafting equipment; and
      (4) References.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall Advise Command on the communication suite, IAW references, by:

   a. Monitoring the External Communications System; and

   b. (I0160) Monitoring the Internal Communications System.

4. Time Required

   a. EO - 20 x 50 min periods;

   b. EC - 3 x 50 min periods (0.5 period / trainee); and

   c. PC - 6 x 50 min periods (1 period / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids

   a. Audio / visual aids; and

   b. System handouts.

7. Test Details. EO 006.01 will be assessed in EC 006.01 and PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development;

b. Self-study to include reviewing equipment handouts and readings publications in order to prepare for lessons; and

c. The Component Overview table summarizes the information that must be trained IAW each component listed in para 10. Instructors must use this table as the basis for training all systems/components in EO 006.01 – 006.05, where the CSEC/O must be proficient in each of the systems in order to monitor them on the job.

COMPONENT OVERVIEW

	Major Points	Amplifying Information
1.	Characteristics / Components	- Components to cabinet level - Key system components
2	Capabilities and limitations	- Casualty modes / redundancy - System common failures - Recovery times - Environmental (Ship's systems)
3	Power distribution	Critical points of failure degrading overall operational capability
4	Manning considerations	
5	Safety requirements for maintenance and operations IAW orders, regulations and technical instructions	
6	Tactical employment / integration of systems within this suite	- Critical points of failure degrading overall operational capability - Suite block diagram

10. Teaching Points

	Description	Instructional Methods	Time	References
TP1	Monitor the External Communications System	Interactive lecture	1 x 50 mins	A21

	(L1050) Monitor Cryptographic Equipment	Guided discussion Self-study		
TP2	(L0140) Monitor the Message Handling and Processing System - (L0150) Monitor the Red/Black Message Sub System		1 x 50 mins	A14
TP3	(J0180) Monitor the Very High Frequency (VHF) System - (K0210) Monitor the Radio Antenna System		2 x 50 mins	A25
TP4	(J0170) Monitor the Ultra High Frequency (UHF) System - (J0190) Monitor an Air-Ground-Air / Maritime Rear Link (AGA/MRL) System		3 x 50 mins	A22, A23
TP5	(J0220) / (J0210) Monitor a Low Frequency (LF) / Very Low Frequency (VLF) System and Medium Frequency (MF) / High Frequency (HF) System Transmit / Receive		6 x 50 mins	A11, A15, A16, A18, A19, A20, A24
TP6	(O0140) Monitor a Global Command & Control System Maritime (GCCS-M) / Link System		1 x 50 mins	A40, A41
TP7	(K1070) Monitor an Information Screening and Distribution System (ISDS) / Submarine Internet Protocol (Sub IP)		0.5 x 50 mins	C4, C5
TP8	(I0150) Monitor a Submarine Escape Communications System		1 x 50 mins	A13
TP9	(I0160) Monitor an Internal Communications System (P0180) Monitor an Event Recorder System		0.5 x 50 mins	A26
TP10	(I1100) Monitor a Sound Powered Network		0.5 x 50 mins	A26

TP11	(I1040) Monitor a Victoria-class Communications System (VCS) 446		0.5 x 50 mins	A26
TP12	(I0980) Monitor a Voice Internal Communications Submarine (VICES)		2 x 50 mins	A26
TP13	(Q0150) Monitor a Training Simulator		1 x 50 mins	N/A – IAW Trainer SOP's

Enabling Check - EO 006.01 - Advise Command on the Communication Suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.01 will be a practical assessment based on content from EO 006.01.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.01 will be a practical assessment based on content from EO 006.01.

2. During this practical assessment, you will be required to advise Command on the communication suite. You must pass this assessment to pass EO 006.01.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have 0.5 x 50 min period to present this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical EC. Trainees to present produced diagrams to the instructor in the form of a brief, to include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented. Instructor is to review drawings prior to EC IAW with Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Comm Suite brief, to include:			
Characteristics / components			
Capabilities and limitations			

Performance Factors	SAT	UNSAT	Comments
Power distribution			
Safety considerations			
Environmental (ship’s systems)			
Casualty modes / redundancy			
Common system failures			

The trainee must achieve 5 of 7 SAT to pass this EC.

PO 006 Advise Command on the CS Suite

EO 006.02 Advise Command on the Electronic Warfare (EW) Suite

1. Performance. Advise Command on the EW suite.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;

      (2) Engineering drafting equipment;

      (3) Supervision; and

      (4) References.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall advise Command on the EW suite, IAW references, by:

   a. (E1280) Monitoring an EW system;

   b. (M0150) Monitoring an AIS; and

   c. (O0170) Monitoring an IFF System.

4. Time Required

   a. EO - 2 x 50 min periods;

   b. EC - 3 x 50 min periods (0.5 period / trainee); and

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids

   a. Audio / visual aids; and

   b. System handouts.

7. Test Details. EO 006.02 will be assessed in EC 006.02 and PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development;

b. Self-study to include reviewing equipment handouts and reading publications in order to prepare for lessons; and

c. The Component Overview table summarizes the information that must be trained IAW each component listed in para 10. Instructors must use this table as the basis for training all systems/components in EO 006.01 – 006.05, where the CSEC/O must be proficient in each of the systems in order to monitor them on the job.

COMPONENT OVERVIEW

	Major Points	Amplifying Information
1	Characteristics / Components	- Components to cabinet level - Key system components
2	Capabilities and limitations	- Casualty modes / redundancy - System common failures - Recovery times - Environmental (Ship's systems)
3	Power distribution	Critical points of failure degrading overall operational capability
4	Manning considerations	
5	Safety requirements for maintenance and operations IAW orders, regulations and technical instructions	
6	Tactical employment / integration of systems within this suite	- Critical points of failure degrading overall operational capability - Suite block diagram

10. Teaching Points

	Description	Instructional Method	Time	References

TP1	(E1280) Monitor an EW system	Interactive lecture Guided discussion Self-study	1 x 50 mins	A36
TP2	(M0150) Monitor an AIS System		0.5 x 50 mins	A35
TP3	(O0170) Monitor an IFF System		0.5 x 50 mins	A37

Enabling Check - EO 006.02 - Advise Command on the EW suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.02 will be a practical assessment based on content from EO 006.02.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.02 will be a practical assessment based on content from EO 006.02.

2. During this practical assessment, you will be required to advise Command on the EW suite. You must pass this assessment to pass EO 006.02.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have a 0.5 x 50 min period to present this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Trainees shall present diagrams produced to the instructor in the form of a brief, and must include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented individually. Instructor is to review drawings prior to EC IAW Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
EW Suite brief, to include:			
Characteristics /components			
Capabilities and limitations			
Power distribution			

Performance Factors	SAT	UNSAT	Comments
Safety considerations			
Environmental (ship's systems)			
Casualty modes / redundancy			
Common system failures			

The trainee must achieve 5 of 7 SAT to pass this EC.

PO 006 Advise Command on the CS Suite

EO 006.03 Advise Command on the Navigation Suite

1. Performance. Advise Command on the navigation suite.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;
      (2) Engineering drafting equipment;
      (3) Supervision; and
      (4) References.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall advise Command on the navigation suite, IAW references, by:

   a. (N0130) Monitoring a Depth Sounder System;

   b. (N0140) Monitoring a Doppler Speed Log System;

   c. (M0180) Monitoring an Inertial Navigation System;

   d. (M0160) Monitoring a Navigation Radar System;

   e. (M1210) Monitoring a Navigation Data Distribution System (NDDS);

   f. (N0170) Monitoring a Shipboard Integrated Navigation And Display System (SHINNADS);

   g. (H0190) Monitoring a Bathy System; and

   h. (F0250) Monitoring a Submarine Periscope System.

4. Time Required

   a. EO - 12 x 50 min periods; and

   b. EC - 3 x 50 min periods (0.5 period / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids

a. Audio / visual aids; and

b. System handouts.

7. Test Details. EO 006.03 will be assessed in EC 006.03 and PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development;

b. Self-study to include reviewing equipment handouts and reading publications in order to prepare for lessons; and

c. The Component Overview table summarizes the information that must be trained IAW each component listed in para 10. Instructors must use this table as the basis for training all systems/components in EO 006.01 – 006.05, where the CSEC/O must be proficient in each of the systems in order to monitor them on the job.

COMPONENT OVERVIEW

	Major Points	Amplifying Information
1.	Characteristics / Components	- Components to cabinet level - Key system components
2	Capabilities and limitations	- Casualty modes / redundancy - System common failures - Recovery times - Environmental (Ship's systems)
3	Power distribution	Critical points of failure degrading overall operational capability
4	Manning considerations	
5	Safety requirements for maintenance and operations IAW orders, regulations and technical instructions	
6	Tactical employment / integration of systems within this suite	- Critical points of failure degrading overall operational capability - Suite block diagram

10. Teaching Points

	Description	Instructional Methods	Time	References
TP1	(N0130) Monitor a Depth Sounder System - Shallow echo sounder - Deep echo sounder	Interactive lecture Guided discussion Self-study	1 x 50 mins	A30, A32
TP2	(N0140) Monitor a Speed Log System		1 x 50 mins	A31
TP3	Monitor a depth system		1 x 50 mins	A5
TP4	(M0180) Monitor an Inertial Navigation System		2 x 50 mins	A28
TP5	(M1210) Monitor a NDDS - (M0130) Monitor a GPS system - (N1210) Monitor a time server		2 x 50 mins	A29, A33
TP6	(N0170) Monitor a SHINNADS system		1 x 50 mins	A34
TP7	(H0190) Monitor a Bathy System - Mk 8F Bathythermograph - 2004 Bathythermograph		1 x 50 mins	A71
TP8	(F0250) Monitor a Submarine Periscope System - (S1430) Initialize Tactical Television System to an operational state - (S1680) Monitor a Tactical Television System		2 x 50 mins	A42, A43, A44
TP9	(M0160) Monitor a Navigation Radar System - (S1720) Monitor a Desiccant System - Fitted Radar System - Portable Radar System		1 x 50 mins	A38, A39, A12

Enabling Check - EO 006.03 - Advise Command on the Navigation Suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.03 will be a practical assessment based on content from EO 006.03.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.03 will be a practical assessment based on content from EO 006.03.

2. During this practical assessment, you will be required to advise Command on navigation suite. You must pass this assessment to pass EO 006.03.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have a 0.5 x 50 min period to present this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Trainees to present produced diagrams to the instructor in the form of a brief, to include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented. Instructor is to review drawings prior to EC IAW with Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Navigation Suite brief, to include:			
Characteristics /components			
Capabilities and limitations			

Performance Factors	SAT	UNSAT	Comments
Power distribution			
Safety considerations			
Environmental (ship's systems)			
Casualty modes / redundancy			
Common system failures			

The trainee must achieve 5 of 7 SAT to pass this EC.

PO 006 Advise Command on the CS Suite

EO 006.04 Advise Command on the Sonar Suite

1. Performance. Advise Command on the sonar suite.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;

      (2) Engineering drafting equipment;

      (3) Supervision; and

      (4) References.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall advise Command on the sonar suite, IAW references, by:

   a. (H1310) Monitoring a Speed Dependent Tonal System;

   b. (H1250) Monitoring a Cavitation Indicator System;

   c. (H1190) Monitoring a Augmenter System;

   d. (H1130) Monitoring an Emergency Underwater Telephone (EUWT) System;

   e. (H0230) Monitoring an Underwater Telephone (UWT) System; and

   f. (G1380) Monitoring a Main Sonar System.

4. Time Required

   a. EO - 7 x 50 min periods; and

   b. EC - 3 x 50 min periods (0.5 / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids

a. Audio / visual aids; and

b. System handouts.

7. Test Details. EO 006.04 will be assessed in EC 006.04 and PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development;

b. Self-study to include reviewing equipment handouts and reading publications in order to prepare for lessons; and

c. The Component Overview table summarizes the information that must be trained IAW each component listed in para 10. Instructors must use this table as the basis for training all systems/components in EO 006.01 – 006.05, where the CSEC/O must be proficient in each of the systems in order to monitor them on the job.

COMPONENT OVERVIEW

	Major Points	Amplifying Information
1.	Characteristics / Components	- Components to cabinet level - Key system components
2	Capabilities and limitations	- Casualty modes / redundancy - System common failures - Recovery times - Environmental (Ship's systems)
3	Power distribution	Critical points of failure degrading overall operational capability
4	Manning considerations	
5	Safety requirements for maintenance and operations IAW orders, regulations and technical instructions	
6	Tactical employment / integration of systems within this suite	- Critical points of failure degrading overall operational capability - Suite block diagram

10. Teaching Points

	Description	Instructional Methods	Time	References
TP1	(H1310) Monitor a Speed Dependent Tonal System	Interactive lecture Guided discussion Self-study	0.5 x 50 mins	A69
TP2	(H1250) Monitor a Cavitation Indicator System		0.5 x 50 mins	A57
TP3	(H1190) Monitor a Augmenter System		0.5 x 50 mins	A66
TP4	(H1130) Monitor an EUWT System		0.5 x 50 mins	A65
TP5	(H0230) Monitor an UWT System		1 x 50 mins	A64
TP6	(G1380) Monitor a Main Sonar System - (G1440) Monitor a 2046 sonar - Passive Ranging - Active Intercept and Ranging - Bow array - Flank array - Towed array		4 x 50 mins	A63, A68, A70

Enabling Check - EO 006.04 - Advise Command on the Sonar suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.04 will be a practical assessment based on content from EO 006.04.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.04 will be a practical assessment based on content from EO 006.04.

2. During this practical assessment, you will be required to advise Command on the sonar suite. You must pass this assessment to pass EO 006.04.

3. References will be provided.

4. Pass mark is 5 of 7 SAT.

5. You will have a 0.5 x 50 min period to present this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Trainees to present produced diagrams to the instructor in the form of a brief, to include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented. Instructor is to review drawings prior to EC IAW Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Sonar Suite brief, to include:			
Characteristics / components			
Capabilities and limitations			

Performance Factors	SAT	UNSAT	Comments
Power distribution			
Safety considerations			
Environmental (ship's systems)			
Casualty modes / redundancy			
Common system failures			

The trainee must achieve 5 of 7 SAT to pass this EC.

PO 006 Advise Command on the CS Suite

EO 006.05 Advise Command on the Fire Control Suite

1. Performance. Advise Command on the fire control suite.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;

      (2) Engineering drafting equipment;

      (3) Supervision; and

      (4) References.

   b. Denied: Assistance.

   c. Environmental: Classroom.

3. Standard. The trainee shall advise Command on the fire control suite, IAW references, by:

   a. (F0260) Monitoring a Torpedo Handling System;

   b. (F0270) Monitoring a Torpedo Launching System;

   c. (H0210) Monitoring a Submerged Signal Ejector (SSE) System; and

   d. (F0280) Monitoring SFCS.

4. Time Required

   a. EO - 15 x 50 min periods; and

   b. EC - 3 x 50 min periods (0.5 / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Guided discussion; and

   c. Self-study.

6. Training Aids

   a. Audio / visual aids; and

   b. System handouts.

7. Test Details. EO 006.05 will be assessed in EC 006.05 and PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development;

b. Self-study to include reviewing equipment handouts and reading publications in order to prepare for lessons; and

c. The Component Overview table summarizes the information that must be trained IAW each component listed in para 10. Instructors must use this table as the basis for training all systems/components in EO 006.01 – 006.05, where the CSEC/O must be proficient in each of the systems in order to monitor them on the job.

COMPONENT OVERVIEW

	Major Points	Amplifying Information
1	Characteristics / Components	- Components to cabinet level - Key system components
2	Capabilities and limitations	- Casualty modes / redundancy - System common failures - Recovery times - Environmental (Ship's systems)
3	Power distribution	Critical points of failure degrading overall operational capability
4	Manning considerations	
5	Safety requirements for maintenance and operations IAW orders, regulations and technical instructions	
6	Tactical employment / integration of systems within this suite	- Critical points of failure degrading overall operational capability - Suite block diagram

10. Teaching Points

	Description	Instructional Method	Time	References
TP1	(H0210) Monitor a SSE System - First level system - (S1670) Managing SSE stores	Interactive lecture Guided discussion Self-study	2 x 50 mins	A8, A53, A54, A55, A56
TP2	(F0260) Monitor a Torpedo Handling System - FMF specialty testing equipment - Minimum Tube Total Loading (MTTL)		3 x 50 mins	A50, A53, A54, A55
TP3	(F0270) Monitor a Torpedo Launching System - First level system - Weapon Discharge Hydraulic System - Air Turbine Pump (ATP) - Torpedo tubes and fittings (water-tight integrity) - Latent Failure Checks (LFC)		7 x 50 mins	A51, A52, A53, A54, A55
TP4	(F0280) Monitor SFCS		3 x 50 mins	A48, A49, A53, A54, A55, A67

Enabling Check - EO 006.05 - Advise Command on the Fire Control Suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.05 will be a practical assessment based on content from EO 006.05.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 006.05 will be a practical assessment based on content from EO 006.05.

7. During this practical assessment, you will be required to advise Command on the fire control suite. You must pass this assessment to pass EO 006.05.

8. References will be provided.

9. Pass mark is 5 of 7 SAT.

10. You will have a 0.5 x 50 min period to present this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical EC. Trainees to present produced diagrams to the instructor in the form of a brief, to include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented. Instructor is to review drawings prior to EC IAW with Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Fire Control (FC) Suite brief, to include:			
Characteristics /components			
Capabilities and limitations			

Performance Factors	SAT	UNSAT	Comments
Power distribution			
Safety considerations			
Environmental (ship’s systems)			
Casualty modes / redundancy			
Common system failures			

The trainee must achieve 5 of 7 SAT to pass this EC.

PO 006 Advise Command on the CS Suite

EO 006.06 Advise Command on CS Suite Integration

1. Performance. Advise Command on CS Suite Integration.

2. Conditions

   a. Given:

      (1) Laptop / Workstation;

      (2) Engineering drafting equipment; and

      (3) References.

   b. Denied: Supervision.

   c. Environmental: Classroom.

3. Standard. The trainee shall advise Command on CS suite integration, IAW references, by:

   a. Monitoring CS network protocols;

   b. Monitoring CS suite; and

   c. Briefing Command.

4. Time Required

   a. EO - 10 x 50 min periods; and

   b. EC - 3 x 50 min periods (0.5 period / trainee).

5. Instructional Methods

   a. Interactive lecture;

   b. Study assignment;

   c. Self-study;

   d. Case study; and

   e. Guided discussion.

6. Training Aids

   a. Audio / visual aids.

7. Test Details. EO 006.06 will be assessed in EC 006.06 as a brief on the interconnectivity of the CS suites, and in PC 006.

8. References. See para 10.

9. Remarks

a. Recommend this EO for DL development.

b. TP2 - Trainee will generate a block diagram of the complete tactical weapons suite to include materials covered in EO 006.01 – 05.

10. Teaching Points

	Description	Instructional Method	Time	Reference
TP1	Monitor CS Network Protocols - (H0220) Monitor a Submarine WSDB - IP addressing and network architecture - Monitor CS LAN systems	Interactive lecture Self-study	1 x 50 mins	A67, A27
TP2	Monitor CS suite - CS Suite Block Diagram - Tactical employment of the CS suite	Interactive lecture Take-home assignment Case study	7 x 50 mins	A79

TP3	Briefing Command
	- Situation
	- Possible causes
	- Engineering and operational impact
	- Options to deal with the situation with details:
	  o Work-arounds
	  o Repair way ahead
	  o Limitations
	  o Supply considerations
	  o Timeline of repair
	- Recommendation
	- Implementation of plan
	- SMEs and external agencies to liaise with or require support from
	- Administration and documentation requirement
		Guided discussion	2 x 50 mins	A2

Enabling Check - EO 006.06 - Advise Command on CS Suite Integration

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.06 will be a practical assessment based on content from EO 006.06.

3. Pass mark will be 5 of 7 SAT.

4. Time required: 3 x 50 min periods (0.5 period / trainee).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 006.06 will be a practical assessment based on content from EO 006.06.

7. During this practical assessment, you will be required to advise Command on CS Suite Integration. You must pass this assessment to pass EO 006.06.

8. References will be provided.

9. Pass mark is 5 of 7 SAT.

10. You will have a 0.5 x 50 min period to present this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Practical EC. Trainees to present produced diagrams to the instructor in the form of a brief, to include the performance factors contained in the table below. Trainees are to be sequestered while diagrams are being presented. Instructor is to review drawings prior to EC IAW with Annex L.

PART IV – ENABLING CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Components required to fire Mk 48			
Interconnections			

Performance Factors	SAT	UNSAT	Comments
SFCS to TOC			
SFCS to weapon			
Navigation Suite to SFCS			
SHINNADS Dual Monitor (SDM) inputs			
SFCS inputs			
Suite interconnections (EO 006.01- 05)			

The trainee must achieve 5 of 7 SAT to pass this EC.

PERFORMANCE CHECK - PO 006 - Advise Command on the CS Suite

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 006 will be a practical assessment based on content from PO 006.

3. Pass mark will be 9 of 12 SAT.

4. Time required: 6 x 50 min periods (1 period / trainee).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 006 will be a practical assessment based on content from PO 006.

2. During this practical assessment, you will be required to advise Command on the CS suite. You must pass this assessment to pass PO 006.

3. References will be provided.

4. Pass mark is 9 of 12 SAT.

5. You will have 1 x 50 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

1. Scenario-based assessment on the CS suite while the submarine is at sea. Scenario will contain one or more faults, where trainees will apply training give an initial brief to Command. Trainee will then be given updates about the situation, and must adapt and incorporate this new information in preparing/presenting a full brief. Instructor will role-play as Command. All parts of brief must be logical to the scenario.

PART IV – PERFORMANCE CHECK LIST

Performance Factors	SAT	UNSAT	Comments
Situation			

Performance Factors	SAT	UNSAT	Comments
Possible causes			
Engineering and operational impact			
Detail Courses of Action (CoA)			
Work-arounds			
Repair way ahead			
Limitations			
Supply considerations			
Timeline for repair			
Recommendations			
Implementation of plan			
SMEs and external agencies to liaise with or require support from			
Administration and documentation requirement			

The trainee must achieve 9 of 12 SAT to pass this PC.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	A-GG-040-001/AG-001 (General Safety Programme Vol. 1)
A2	B-GJ-005-500/FP-000 (Canadian Forces Operational Planning Process (OPP))
A3	B-GN-002-000/RQ-001 (CFCD 102 - RCN Combat Readiness / Training Requirements)
A4	B-GN-005-RCN/RQ-001(CFCD 129 - RCN Readiness & Sustainment Policy)
A5	BR1966(19) (Ship's Depth System)
A6	C-03-005-012/AM-012 (Naval Material Management System Manual (NaMMS))
A7	C-03-005-033/AA-000 (Naval Engineering Manual (NEM))
A8	C-09-153-003/TS-000 (Explosives Safety Manual Vol. 3)
A9	C-23-SUB-002/MS-001 (CFCD 103 - Submarine Escape and Rescue Manual)
A10	C-23-VIC-000/AG-001 (MATCERT – Victoria Class)
A11	C-27-C48-B00/MS001 (Handbook and Parts List (Weapons) Integrated Communications Mast Outfit WHC)
A12	C-27-D09-000/MS-001 (Desiccation Equipment)
A13	C-51-701-000/MS-001 (Expendable Communications Buoy - Outfit)
A14	C-51-832-000/MS-001 (Message Handling & Distribution System)
A15	C-51-834-000/MS-001 (Aerial Exchange Unit & Indicator Unit)
A16	C-51-836-000/MS-001 (Buoyant Wire Antenna Group)
A17	C-51-837-000/MS-001 (VLF/LF Submarine Broadcast Crypto)
A18	C-51-838-A00/MS-001 (Handbook for Transceiver Type H4640)
A19	C-51-838-B00/MS-001 (Handbook for 250W HF Amplifier Type H1071)
A20	C-51-839-000/MS-001 (VLF/LF MSK Receiver Outfit CFD)
A21	C-51-846-000/MS-001 (Tactical Crypto Equipment)
A22	C-51-853-000/MS-001 (UHF Secure Voice System)
A23	C-51-883-000 /MS-001 (UHF SATCOM Mini-DAMA System)
A24	C-51-916-000/MS-001 (Handbook for Submarine Communications and Control System Outfits SSC/KMW/EUC (BR8111))

Ref Code	Publication
A25	C-51-919-000/MS-001 (Instruction Manual VHF Marine Transceiver IC-M601)
A26	C-52-453-000/MS-001 (Rationalized Internal Communications)
A27	C-57-786-000/MS-001 (RS422 Data Distribution Unit)
A28	C-57-794-000/MS-001 (Navigation System, Ring Laser Gyro Navigator)
A29	C-57-796-000/MS-001 (Naval GPS Navigation (NGN III) System)
A30	C-57-803-000/MS-001 (Echo Sounder Type 778JJ)
A31	C-57-808-000/MS-001 (Ships Log Outfit SSL AGILOG)
A32	C-57-809-000/MS-001 (Echo Sounder Type 780EH)
A33	C-57-852-000/MS-001 (Navigation Data Distribution System)
A34	C-57-856-000/MS-001 (SHINNADS Table Console)
A35	C-57-860-000/MS-001 (Automatic Identification System)
A36	C-58-407-000/MS-001 (Maintenance Manual CP-5136/ULR 501 Data Processor AN/ULR 501)
A37	C-59-678-000/MS-001 (IFF System Group, Mk. 12)
A38	C-59-875-000/MS-001 (Radar Set, Navigation, KH 1007)
A39	C-59-952-000/MS-001 (Radar Set, Navigation, FURUNO)
A40	C-60-206-000/MS-001 (Operating & Maintenance Manual Global Command & Control System - Maritime TAC-3 Processor OJ-683A(V)RCN)
A41	C-60-211-001/MS-001 (Technical Manual Operation and Maintenance Instructions CP-2205 (P) (V) / USQ-125)
A42	C-66-382-000/MS-001 (Tactical Television Console (TTVC))
A43	C-66-383-000/MS-001 (System, Search Periscope)
A44	C-66-384-000/MS-001 (System, Attack Periscope)
A45	C-69-008-000/TS001 (Safety Storage and Handling Instructions Otto Fuel II)
A46	C-69-726-000/FP-001 (Submarine Torpedo Reference Manual Foreign Military Sales Volume 1) SECRET
A47	C-69-726-000/FP-002 (Submarine Torpedo Reference Manual Foreign Military Sales Volume 2) SECRET
A48	C-69-815-000/MB-001(SFCS Operators Manual)
A49	C-69-815-000/MS-002 (Interactive Electronic Technical Manual (IETM) BSSU Interface Victoria Class Submarine CCS 876)
A50	C-69-817-000/MS-001 (WHS Technical Description)
A51	C-69-818-000/MS-004 (Weapons Discharge System Technical Description)
A52	C-69-818-000/MS-006 (ATP Technical Description and Maintenance)
A53	C-69-820-000/MB-006 (SUBWOPS Vol. 1)

Ref Code	Publication
A54	C-69-820-000/MB-006 (SUBWOPS Vol. 2)
A55	C-69-820-000/MB-006 (SUBWOPS Vol. 3)
A56	C-69-821-000/MS-001 (SSE Handbook)
A57	C-69-823-000/MS-001 (Sonar Set 189BD)
A58	C-69-827-000/MS-001 (Handbook and Part List (Acoustic) 2046 Sonar Outboard)
A59	C-69-827-000/NG-003 (Test Sheet Shipboard Systems and Equipment 2046 Sonar Outboard Towed Array Functional Check)
A60	C-69-827-000/NK-001 (Naval PM Performance Test (Acoustics) 2046 Sonar Pre-Deployment Checks)
A61	C-69-827-000/NK-002 (Naval PM Performance Test (Acoustics) 2046 Sonar Post-Deployment Checks)
A62	C-69-827-000/NK-003 (Naval PM Schedule (Acoustics) 2046 Sonar Outboard System Confidence Check
A63	C-69-838-000/MS-001 (Towed Array Systems)
A64	C-69-846-000/MS-001 (Sonar Type 2008 Under Water Telephone, Variant NG)
A65	C-69-860-000/MS-001 (Emergency Underwater Telephone 2073)
A66	C-69-866-000/MS-001 (Sonar 2071 ADA Sub-Acoustic Augmenter)
A67	C-69-867-000/MS-001 (Weapons System Databus (Generic) Handbook)
A68	C-69-887-001/MS-001 (AN/BQQ-10V7 Sonar Suite TI-10)
A69	C-69-888-000/MS-001 (Speed Dependent Tonal Array System)
A70	C-69-891-000/MS-001 (Passive Ranging Sonar)
A71	C-69-894-000/MS-001 (MK8-F Bathythermograph System)
A72	C-77-360-000/MV-001 (Damage Control Information HMCS VICTORIA – Electrical Engineering Systems)
A73	C-77-360-000/MV-005 (Damage Control Information HMCS VICTORIA – Mechanical Systems Engineering)
A74	C-77-360-000/MV-009 (Damage Control Information HMCS VICTORIA – Combat Systems Engineering)
A75	C-77-360-000/MV-017 (Damage Control Information HMCS VICTORIA – Command Briefing Cards
A76	C-77-360-000/MV-021 (Damage Control Information HMCS VICTORIA – Compartment Kill Cards)
A77	C-77-360-000/MV-025 (Damage Control Information HMCS VICTORIA – Loss of Load Centers LC1 & LC2)
A78	C-77-360-000/MV-029 (Damage Control Information HMCS VICTORIA – Change-Over Switches)

Ref Code	Publication
A79	C-77-360-000/MV-10 (Damage Control Information HMCS Windsor CSE)
A80	Canadian Submarine Attack Drills (CANSADs)
A81	CFCD 122 (Victoria Class Submarine Weapon Certification Programme)
A82	National Defence Security Orders and Directives (NDSOD) Chapter 5 &15
A83	National Defence Security Orders and Directives (NDSOD) Physical Security Technical Standard Manual
A84	NAVORD 1150-2 (SUBSAFE Licensing and Certification Policy)
A85	NAVORD 2130-6 (Security of Small Arms and Small Arms Ammunition)
A86	NAVORD 3136-4 (Weapons Keys for Ships and Submarines)
A87	NAVORD 3250-7 (RCN Material OPDEF Process)
A88	NAVORD 4995-0 (Safety Firing Orders for Ships and Submarines)
A89	Occupation Military Employment Structure / Job-based specification
A90	Safety and Environment Management System (SEMS) Manual
A91	Victoria Class Emergency Operating Procedures (VCEOP)
A92	Victoria Class Ship's Standing Orders (VCSSO)
A93	Victoria Class Standard Operating Procedures (VCSOP)
A94	Victoria Class Submarine Work-ups Guide (VCSWG)
C1	National Institute for Occupational Safety and Health (NIOSH) Pocket Guide to Chemical Hazards
C2	Emergency Response Guidebook
C3	Workplace Hazardous Materials Information System (WHMIS) Handouts
C4	Sub IP System Operator and Maintainer Instructor Guide
C5	ISDS Student Guide
C6	Project Management Body of Knowledge (PMBOK) http://www.pmi.org/PMBOK-Guide-and-Standards.aspx

ANNEX B – TASK LIST

Task No	Task Statement	Train	PO/EO
A0040	Resolve an FMF assistance problem	Yes	003.02
A0120	Draft a Planned Maintenance (PM) schedule	Yes	003.02
A0190	Draft an Operational Deficiency Message (OPDEF)	Yes	003.02
A0210	Evaluate/Interpret Pre-Sail checks reports	Yes	003.03
A0450	Manage CSE department pre-sail checks	Yes	003.03
A0470	Manage the CSE department in the absence of a HOD qualified CSEO	Yes	002.03
A0620	Provide a recommendation to Command on combat systems equipment availability and reconfiguration options	Yes	005.01
A0640	Provide input to a Periodic Engineering letter	Yes	003.01
A0830	Track and report work period progress	Yes	003.02
A0840	Track CS Combat Readiness Requirements (CRR) progress	Yes	002.03
A0860	Analyze/evaluate/interpret a CS suite level test/trial/inspection result or report	Yes	003.02
A0870	Brief Operations Room Officer (ORO) on CS equipment status	Yes	
A0990	Coordinate departmental CM activities	Yes	003.02
A1000	Coordinate departmental PM activities	Yes	003.02
A1040	Coordinate ship-level maintenance activities	Yes	003.02
A1080	Diagnose CS suite-level faults	Yes	005.01
A1110	Analyze/evaluate/interpret a ship-level test/trial/inspection result or report	Yes	003.02
A1160	Direct W ENG Sonar pre-sail checks	Yes	003.03
A1170	Direct W ENG Communication pre-sail checks	Yes	003.03
A1200	Direct W ENG Fire Control pre-sail checks	Yes	003.03
A1210	Coordinate divisional technical readiness	Yes	
A1220	Managing technical administration	Yes	003.01
A1230	Direct the conduct of weapons systems engineering preventive maintenance	Yes	003.02
A1240	Direct the repair of weapons systems engineering faults	Yes	003.02
A1250	Coordinating system trials	Yes	003.02
A1260	Maintain configuration management	Yes	003.01
A1270	Provide options/recommendations on weapons systems engineering corrective maintenance to Command	Yes	003.02
A1280	Conduct risk assessment (IAW C-23-….)	Yes	003.05
A1290	Evaluate weapons systems engineering faults	Yes	003.02
A1300	Coordinate the conduct of weapons systems engineering test forms and trials	Yes	003.02
A1310	Support WCP	Yes	
A1320	Supporting WCP administration	Yes	004.01
A1330	Advise Command on CS suite-level equipment	Yes	003.02
B0030	Investigate/review a CF 98	Yes	002.02
B0040	Classify/identify HAZMAT	Yes	001.02

Task No	Task Statement	Train	PO/EO
B0100	Draft a General Safety Accident Prevention Report (CF 663)	Yes	002.02
B0120	Draft a Report on Injuries or Exposure to Toxic Material (CF 98)	Yes	002.02
B0180	Enforce HAZMAT storage compatibility regulations	Yes	001.01
B0250	Investigate/review a CF 663	Yes	002.02
B0270	Maintain a missile or torpedo log	Yes	004.02
B0350	Operate HAZMAT detection equipment (e.g. OTTO fuel detector, PHD Ultra)	Yes	001.02
B0740	Conduct a Hazardous Material (HAZMAT) inspection	Yes	001.01
B0860	Direct a CS Emergency Response Team (ERT)	Yes	005.01
B0890	Direct response to a HAZMAT incident	Yes	001.02
B0900	Coordinate unit HAZMAT responsibilities	Yes	PO 001
B0910	Managing unit HAZMAT program readiness	Yes	001.01
B0920	Manage W ENG department safety IAW SEMS	Yes	002.02
B0930	Direct the maintenance equipment associated with HAZMAT clean up	Yes	001.01
E1280	Monitor a SeaSearch II	Yes	006.02
E1290	Direct a SeaSearch II functional check	Yes	003.02
E1300	Direct a SeaSearch II corrective maintenance	Yes	003.02
E1310	Direct a SeaSearch II preventive maintenance	Yes	003.02
E1320	Analyze a SeaSearch II functional check result	Yes	003.02
E1330	Analyze a SeaSearch II test/trial/inspection result or report	Yes	003.02
F0020	Embark/disembark a torpedo	Yes	004.02
F0030	Fire a torpedo in emergency mode	Yes	004.03
F0040	Fire a torpedo in primary mode (e.g. OPS Room/Control Room)	Yes	004.03
F0090	Analyze/evaluate/interpret a Torpedo Handling System functional check result	Yes	003.02
F0120	Analyze/evaluate/interpret a Torpedo Handling System test/trial/inspection result or report	Yes	003.02
F0130	Analyze/evaluate/interpret a Torpedo Launching System functional check result	Yes	003.02
F0140	Analyze/evaluate/interpret a Torpedo Launching System test/trial/inspection result or report	Yes	003.02
F0150	Inspect/test/connect an Control Cable	Yes	004.02
F0250	Monitor a Submarine Periscope System's performance	Yes	006.03
F0260	Monitor a Torpedo Handling System performance	Yes	006.05
F0270	Monitor a Torpedo Launching System performance	Yes	006.05
F0280	Monitor SFCS performance	Yes	006.05
F0740	Supervise the embarking/disembarking of a torpedo	Yes	004.02
F0760	Band / de-band a torpedo	Yes	004.02
F0880	Carry out a Torpedo Launching System misfire routine	Yes	004.03
F1100	Direct a corrective maintenance/repair activity on a SFCS	Yes	003.02
F1110	Direct a corrective maintenance/repair activity on a Submarine Periscope System	Yes	003.02

Task No	Task Statement	Train	PO/EO
F1120	Direct a corrective maintenance/repair activity on a Torpedo Handling System	Yes	003.02
F1130	Direct a corrective maintenance/repair activity on a Torpedo Launching System	Yes	003.02
F1140	Analyze/evaluate/interpret a SFCS functional check result	Yes	003.02
F1150	Analyze/evaluate/interpret a SFCS test/trial/inspection result or report	Yes	003.02
F1190	Direct a SFCS functional check	Yes	003.02
F1200	Direct a SFCS PM activity (including a test/trial/inspection)	Yes	003.02
F1210	Direct a Submarine Periscope System functional check	Yes	003.02
F1220	Direct a Submarine Periscope System PM activity (including a test/trial/inspection)	Yes	003.02
F1230	Direct a Torpedo Handling System functional check	Yes	003.02
F1240	Direct a Torpedo Handling System PM activity (including a test/trial/inspection)	Yes	003.02
F1250	Direct a Torpedo Launching System functional check	Yes	003.02
F1260	Direct a Torpedo Launching System PM activity (including a test/trial/inspection)	Yes	003.02
F1270	Analyze/evaluate/interpret a Submarine Periscope System functional check result	Yes	003.02
F1280	Analyze/evaluate/interpret a Submarine Periscope System test/trial/inspection result or report	Yes	003.02
G1380	Monitor a BQQ/10	Yes	006.04
G1390	Direct a BQQ/10 functional check	Yes	003.02
G1400	Direct a BQQ/10 corrective maintenance	Yes	003.02
G1410	Direct a BQQ/10 preventive maintenance	Yes	003.02
G1420	Analyze a BQQ/10 functional check result	Yes	003.02
G1430	Analyze a BQQ/10 test/trial/inspection result or report	Yes	003.02
G1440	Monitor a 2046 sonar	Yes	006.04
G1450	Direct a 2046 sonar functional check	Yes	003.02
G1460	Direct a 2046 sonar corrective maintenance	Yes	003.02
G1470	Direct a 2046 sonar preventive maintenance	Yes	003.02
G1480	Analyze a 2046 sonar functional check result	Yes	003.02
G1490	Analyze a 2046 sonar test/trial/inspection result or report	Yes	003.02
H0010	Analyze/evaluate/interpret a Submarine WSDB test/trial/inspection result or report	Yes	003.02
H0060	Analyze/evaluate/interpret a Bathy System functional check result	Yes	003.02
H0130	Analyze/evaluate/interpret a Bathy System test/trial/inspection result or report	Yes	003.02
H0190	Monitor a Bathy System performance	Yes	006.03
H0210	Monitor a Submerged Signal Ejector (SSE) System performance	Yes	006.05
H0220	Monitor a Submarine WSDB performance	Yes	006.06
H0230	Monitor an UWT System performance	Yes	006.04

Task No	Task Statement	Train	PO/EO
H0450	Analyze/evaluate/interpret an UWT System test/trial/inspection result or report	Yes	003.02
H0460	Analyze/evaluate/interpret an UWT Systems functional check result	Yes	003.02
H0970	Direct a corrective maintenance/repair activity on an UWT System	Yes	003.02
H0980	Direct an UWT System PM activity (including a test/trial/inspection)	Yes	003.02
H0990	Direct an UWT System functional check	Yes	003.02
H1000	Direct a Bathy System functional check	Yes	003.02
H1010	Direct a Bathy System PM activity (including a test/trial/inspection)	Yes	003.02
H1020	Direct a corrective maintenance/repair activity on a Bathy System	Yes	003.02
H1060	Direct a corrective maintenance/repair activity on a Submarine WSDB	Yes	003.02
H1070	Direct a SSE System functional check	Yes	003.02
H1080	Direct a Submarine WSDB functional check	Yes	003.02
H1090	Direct a Submarine WSDB PM activity (including a test/trial/inspection)	Yes	003.02
H1100	Analyze/evaluate/interpret a SSE System functional check result	Yes	003.02
H1110	Analyze/evaluate/interpret a SSE System test/trial/inspection result or report	Yes	003.02
H1120	Analyze/evaluate/interpret a Submarine WSDB functional check result	Yes	003.02
H1130	Monitor a 2073 Emergency Underwater Telephone (EUWT)	Yes	006.04
H1140	Direct a 2073 EUWT functional check	Yes	003.02
H1150	Direct a 2073 EUWT corrective maintenance	Yes	003.02
H1160	Direct a 2073 EUWT preventive maintenance	Yes	003.02
H1170	Analyze a 2073 EUWT functional check result	Yes	003.02
H1180	Analyze a 2073 EUWT test/trial/inspection result or report	Yes	003.02
H1190	Monitor a 2071 Augmenter	Yes	006.04
H1200	Direct a 2071 Augmenter functional check	Yes	003.02
H1210	Direct a 2071 Augmenter corrective maintenance	Yes	003.02
H1220	Direct a 2071 Augmenter preventive maintenance	Yes	003.02
H1230	Analyze a 2071 Augmenter functional check result	Yes	003.02
H1240	Analyze a 2071 Augmenter test/trial/inspection result or report	Yes	003.02
H1250	Monitor a 189 Cavitation Indicator	Yes	006.04
H1260	Direct a 189 Cavitation Indicator functional check	Yes	003.02
H1270	Direct a 189 Cavitation Indicator corrective maintenance	Yes	003.02
H1280	Direct a 189 Cavitation Indicator preventive maintenance	Yes	003.02
H1290	Analyze a 189 Cavitation Indicator functional check result	Yes	003.02
H1300	Analyze a 189 Cavitation Indicator test/trial/inspection result or report	Yes	003.02
H1310	Monitor a Speed Dependent Tonal system	Yes	006.04
H1320	Direct a Speed Dependent Tonal system functional check	Yes	003.02
H1330	Direct a Speed Dependent Tonal system corrective maintenance	Yes	003.02
H1340	Direct a Speed Dependent Tonal system preventive maintenance	Yes	003.02

Task No	Task Statement	Train	PO/EO
H1350	Analyze a Speed Dependent Tonal system functional check result	Yes	003.02
H1360	Analyze a Speed Dependent Tonal system test/trial/inspection result or report	Yes	003.02
H1380	Direct an SSE corrective maintenance	Yes	003.02
H1390	Direct an SSE preventive maintenance	Yes	003.02
I0150	Monitor a Submarine Escape Communication System performance	Yes	006.01
I0160	Monitor an Internal Communication System performance	Yes	006.01
I0230	Perform a Public Address System functional check	Yes	005.01
I0330	Analyze/evaluate/interpret an Internal Communication System functional check result	Yes	003.02
I0340	Analyze/evaluate/interpret an Internal Communication System test/trial/inspection result or report	Yes	003.02
I0840	Analyze/evaluate/interpret a Submarine Escape Communication System functional check result	Yes	003.02
I0850	Analyze/evaluate/interpret a Submarine Escape Communication System test/trial/inspection result or report	Yes	003.02
I0910	Direct a corrective maintenance/repair activity on an Internal Communication System	Yes	003.02
I0960	Direct an Internal Communication System functional check	Yes	003.02
I0970	Direct an Internal Communication System PM activity (including a test/trial/inspection)	Yes	003.02
I0980	Monitor a VICES	Yes	006.01
I0990	Direct a VICES functional check	Yes	003.02
I1000	Direct a VICES corrective maintenance	Yes	003.02
I1010	Direct a VICES preventive maintenance	Yes	003.02
I1020	Analyze a VICES functional check result	Yes	003.02
I1030	Analyze a VICES test/trial/inspection result or report	Yes	003.02
I1040	Monitor a VCS 446	Yes	006.01
I1050	Direct a VCS 446 functional check	Yes	003.02
I1060	Direct a VCS 446 corrective maintenance	Yes	003.02
I1070	Direct a VCS 446 preventive maintenance	Yes	003.02
I1080	Analyze a VCS 446 functional check result	Yes	003.02
I1090	Analyze a VCS 446 test/trial/inspection result or report	Yes	003.02
I1100	Monitor a Sound Powered Network	Yes	006.01
I1110	Direct a Sound Powered Network functional check	Yes	003.02
I1120	Direct a Sound Powered Network corrective maintenance	Yes	003.02
I1130	Direct a Sound Powered Network preventive maintenance	Yes	003.02
I1140	Analyze a Sound Powered Network functional check result	Yes	003.02
I1150	Analyze a Sound Powered Network test/trial/inspection result or report	Yes	003.02
I1160	Direct a Submarine Escape Communication System functional check result	Yes	003.02
I1170	Direct a Submarine Escape Communication System corrective maintenance	Yes	003.02

Task No	Task Statement	Train	PO/EO
I1180	Direct a Submarine Escape Communication System preventive maintenance	Yes	003.02
J0110	Analyze/evaluate/interpret a UHF System functional check result	Yes	003.02
J0120	Analyze/evaluate/interpret a UHF System test/trial/inspection result or report	Yes	003.02
J0150	Analyze/evaluate/interpret a VHF System test/trial/inspection result or report	Yes	003.02
J0160	Analyze/evaluate/interpret a VHF System test/trial/inspection results/report	Yes	003.02
J0170	Monitor a UHF System performance	Yes	006.01
J0180	Monitor a VHF System performance	Yes	006.01
J0190	Monitor an AGA/MRL System performance	Yes	006.01
J0210	Monitor an HF/MF/LF/VLF Receive System performance	Yes	006.01
J0220	Monitor an HF/MF/LF/VLF Transmit System performance	Yes	006.01
J0330	Analyze/evaluate/interpret an HF/MF/LF/VLF Receive System functional check result	Yes	003.02
J0340	Analyze/evaluate/interpret an HF/MF/LF/VLF Receive System test/trial/inspection result or report	Yes	003.02
J0390	Analyze/evaluate/interpret an HF/MF/LF/VLF Transmit System functional check result	Yes	003.02
J0400	Analyze/evaluate/interpret an HF/MF/LF/VLF Transmit System test/trial/inspection result or report	Yes	003.02
J1090	Direct a corrective maintenance/repair activity on a UHF System	Yes	003.02
J1100	Direct a corrective maintenance/repair activity on a VHF System	Yes	003.02
J1130	Direct a corrective maintenance/repair activity on an HF/MF/LF/VLF Receive System	Yes	003.02
J1140	Direct a corrective maintenance/repair activity on an HF/MF/LF/VLF Transmit System	Yes	003.02
J1150	Direct a UHF System functional check	Yes	003.02
J1160	Direct a UHF System PM activity (including a test/trial/inspection)	Yes	003.02
J1170	Direct a VHF System functional check	Yes	003.02
J1180	Direct a VHF System PM activity (including a test/trial/inspection)	Yes	003.02
J1230	Direct an HF/MF/LF/VLF Receive System functional check	Yes	003.02
J1240	Direct an HF/MF/LF/VLF Receive System PM activity (including a test/trial/inspection)	Yes	003.02
J1250	Direct an HF/MF/LF/VLF Transmit System functional check	Yes	003.02
J1260	Direct an HF/MF/LF/VLF Transmit System PM activity (including a test/trial/inspection)	Yes	003.02
K0180	Analyze/evaluate/interpret a WAN/LAN System test/trial/inspection result or report	Yes	003.02
K0210	Monitor a Radio Antenna System and Structure performance	Yes	006.01
K0880	Analyze/evaluate/interpret a Radio Antenna System and Structure system test/trial/inspection result or report	Yes	003.02
K0910	Analyze/evaluate/interpret a Satellite Communications System functional check result	Yes	003.02

Task No	Task Statement	Train	PO/EO
K0920	Direct a corrective maintenance/repair activity on a Radio Antenna System and Structure	Yes	003.02
K0930	Direct a corrective maintenance/repair activity on a Satellite Communications System	Yes	003.02
K0950	Analyze/evaluate/interpret a Satellite Communications System test/trial/inspection result or report	Yes	003.02
K0960	Direct a corrective maintenance/repair activity on a Radio Antenna System and Structure	Yes	003.02
K0970	Direct a corrective maintenance/repair activity on a WAN/LAN System	Yes	003.02
K1000	Direct a Radio Antenna System and Structure functional check	Yes	003.02
K1010	Direct a Radio Antenna System and Structure PM activity (including a test/trial/inspection)	Yes	003.02
K1020	Analyse WAN/LAN event logs and reports	Yes	003.02
K1030	Direct a Satellite Communications System functional check	Yes	003.02
K1040	Direct a Satellite Communications System PM activity (including a test/trial/inspection)	Yes	003.02
K1050	Direct a WAN/LAN System functional check	Yes	003.02
K1060	Direct a WAN/LAN System PM activity (including a test/trial/inspection)	Yes	003.02
K1070	Monitor a ISDS/Sub IP	Yes	006.01
K1080	Direct an ISDS/Sub IP functional check	Yes	003.02
K1090	Direct an ISDS/Sub IP corrective maintenance	Yes	003.02
K1100	Direct an ISDS/Sub IP preventive maintenance	Yes	003.02
K1110	Analyze an ISDS/Sub IP functional check result	Yes	003.02
K1120	Analyze an ISDS/Sub IP test/trial/inspection result or report	Yes	003.02
L0140	Monitor a Message Processing and Handling System performance	Yes	006.01
L0150	Monitor a Red/Black Message Sub System performance	Yes	006.01
L0660	Analyze/evaluate/interpret a Message Processing and Handling System functional check result	Yes	003.02
L0680	Analyze/evaluate/interpret a Message Processing and Handling System test/trial/inspection result or report	Yes	003.02
L0860	Analyze/evaluate/interpret a Red/Black Message Sub System functional check result	Yes	003.02
L0870	Analyze/evaluate/interpret a Red/Black Message Sub System test/trial/inspection result or report	Yes	003.02
L0940	Direct a corrective maintenance/repair activity on a Message Processing and Handling System	Yes	003.02
L0950	Direct a corrective maintenance/repair activity on a Red/Black Message Sub System	Yes	003.02
L0990	Direct a Message Processing and Handling System functional check	Yes	003.02
L1000	Direct a Message Processing and Handling System PM activity (including a test/trial/inspection)	Yes	003.02

Task No	Task Statement	Train	PO/EO
L1010	Direct a Red/Black Message Sub System functional check	Yes	003.02
L1020	Direct a Red/Black Message Sub System PM activity (including a test/trial/inspection)	Yes	003.02
L1050	Monitor cryptographic equipment	Yes	006.01
L1060	Direct cryptographic equipment functional check	Yes	003.02
L1070	Direct cryptographic equipment corrective maintenance	Yes	003.02
L1080	Direct cryptographic equipment preventive maintenance	Yes	003.02
L1090	Analyze cryptographic equipment functional check result	Yes	003.02
L1100	Analyze cryptographic equipment test/trial/inspection result or report	Yes	003.02
M0130	Monitor a GPS system performance	Yes	006.03
M0150	Monitor an AIS System performance	Yes	006.02
M0160	Monitor a NAV Radar system performance	Yes	006.03
M0180	Monitor an Internal Navigation System performance	Yes	006.03
M0360	Analyze/evaluate/interpret an Internal Navigation System functional check result	Yes	003.02
M0390	Analyze/evaluate/interpret an Internal Navigation System test/trial/inspection result or report	Yes	003.02
M0700	Analyze/evaluate/interpret a GPS functional check result	Yes	003.02
M0710	Analyze/evaluate/interpret a GPS system test/trial/inspection result or report	Yes	003.02
M0800	Analyze/evaluate/interpret an AIS System functional check result	Yes	003.02
M0810	Analyze/evaluate/interpret an AIS System test/trial/inspection result or report	Yes	003.02
M0870	Analyze/evaluate/interpret a NAV Radar system functional check result	Yes	003.02
M0880	Analyze/evaluate/interpret a NAV Radar system test/trial/inspection result or report	Yes	003.02
M1030	Direct a corrective maintenance/repair activity on a GPS system	Yes	003.02
M1050	Direct a corrective maintenance/repair activity on an AIS System	Yes	003.02
M1060	Direct a corrective maintenance/repair activity on an Internal Navigation System	Yes	003.02
M1070	Direct a GPS functional check	Yes	003.02
M1080	Direct a GPS system PM activity (including a test/trial/inspection)	Yes	003.02
M1110	Direct an AIS System functional check	Yes	003.02
M1120	Direct an AIS System PM activity (including a test/trial/inspection)	Yes	003.02
M1130	Direct an Internal Navigation System functional check	Yes	003.02
M1140	Direct an Internal Navigation System PM activity (including a test/trial/inspection)	Yes	003.02
M1150	Direct a corrective maintenance/repair activity on a NAV Radar system	Yes	003.02
M1160	Direct a NAV Radar System functional check	Yes	003.02
M1170	Direct a NAV Radar system PM activity (including a test/trial/inspection)	Yes	003.02

Task No	Task Statement	Train	PO/EO
M1210	Monitor a Navigation Data Distribution System (NDDS)	Yes	006.03
M1220	Direct a NDDS functional check	Yes	003.02
M1230	Direct a NDDS corrective maintenance	Yes	003.02
M1240	Direct a NDDS preventive maintenance	Yes	003.02
M1250	Analyze a NDDS functional check result	Yes	003.02
M1260	Analyze a NDDS test/trial/inspection result or report	Yes	003.02
N0130	Monitor a Depth Sounder System performance	Yes	006.03
N0140	Monitor a Doppler Speed Log System performance	Yes	006.03
N0170	Monitor a SHINNADS system performance	Yes	006.03
N0610	Analyze/evaluate/interpret a Depth Sounder System functional check result	Yes	003.02
N0620	Analyze/evaluate/interpret a Depth Sounder System test/trial/inspection result or report	Yes	003.02
N0670	Analyze/evaluate/interpret a Doppler Speed Log System functional check result	Yes	003.02
N0680	Analyze/evaluate/interpret a Doppler Speed Log System test/trial/inspection result or report	Yes	003.02
N1010	Direct a corrective maintenance/repair activity on a Depth Sounder System	Yes	003.02
N1020	Direct a corrective maintenance/repair activity on a Doppler Speed Log System	Yes	003.02
N1050	Direct a corrective maintenance/repair activity on a SHINNADS system	Yes	003.02
N1070	Direct a Depth Sounder System functional check	Yes	003.02
N1080	Direct a Depth Sounder System PM activity (including a test/trial/inspection)	Yes	003.02
N1090	Direct a Doppler Speed Log System functional check	Yes	003.02
N1100	Direct a Doppler Speed Log System PM activity (including a test/trial/inspection)	Yes	003.02
N1110	Analyze/evaluate/interpret a SHINNADS functional check result	Yes	003.02
N1120	Analyze/evaluate/interpret a SHINNADS system test/trial/inspection result or report	Yes	003.02
N1170	Direct a SHINNADS functional check	Yes	003.02
N1180	Direct a SHINNADS system PM activity (including a test/trial/inspection)	Yes	003.02
N1210	Monitor a time server	Yes	006.03
N1220	Direct a time server functional check result	Yes	003.02
N1230	Direct a time server corrective maintenance	Yes	003.02
N1240	Direct a time server preventive maintenance	Yes	003.02
N1250	Analyze a time server functional check result	Yes	003.02
N1260	Analyze a time server test/trial/inspection result or report	Yes	003.02
O0140	Monitor a JOTS/JMICS/GCCS/EDO LINK System performance	Yes	006.01
O0170	Monitor a IFF System performance	Yes	006.02
O0310	Analyze/evaluate/interpret an IFF System functional check result	Yes	003.02

Task No	Task Statement	Train	PO/EO
O0320	Analyze/evaluate/interpret an IFF System test/trial/inspection result or report	Yes	003.02
O0630	Analyze/evaluate/interpret a JOTS/JMICS/GCCS/EDO LINK System functional check result	Yes	003.02
O0640	Analyze/evaluate/interpret a JOTS/JMICS/GCCS/EDO LINK System test/trial/inspection result or report	Yes	003.02
O0870	Direct a corrective maintenance/repair activity on a JOTS/JMICS/GCCS/EDO LINK System	Yes	003.02
O0890	Direct a corrective maintenance/repair activity on an IFF System	Yes	003.02
O0910	Direct a JOTS/JMICS/GCCS/EDO LINK System functional check	Yes	003.02
O0920	Direct a JOTS/JMICS/GCCS/EDO LINK System PM activity (including a test/trial/inspection)	Yes	003.02
O0990	Direct an Interrogator Friend or Foe (IFF) System functional check	Yes	003.02
O1000	Direct an IFF PM activity (including a test/trial/inspection)	Yes	003.02
P0180	Monitor an Event Recorder System performance	Yes	006.01
P0290	Analyze/evaluate/interpret an Event Recorder System functional check result	Yes	003.02
P0300	Analyze an Event Recorder System test/trial/inspection result or report	Yes	003.02
P0880	Direct a corrective maintenance/repair activity on an Event Recorder System	Yes	003.02
P0960	Direct an Event Recorder System functional check	Yes	003.02
P0970	Direct an Event Recorder System PM activity (including a test/trial/inspection)	Yes	003.02
Q0080	Analyze/evaluate/interpret a Training Simulator functional check result	Yes	003.02
Q0100	Analyze/evaluate/interpret a Training Simulator test/trial/inspection result or report	Yes	003.02
Q0150	Monitor a Training Simulator performance	Yes	003.02
Q0260	Analyze/evaluate/interpret an Emergency Beacon functional check result	Yes	003.02
Q0270	Analyze/evaluate/interpret an Emergency Beacon system test/trial/inspection result or report	Yes	003.02
Q0880	Direct a corrective maintenance/repair activity on a Training Simulator	Yes	003.02
Q0920	Direct a Training Simulator functional check	Yes	003.02
Q0930	Direct a Training Simulator PM activity (including a test/trial/inspection)	Yes	003.02
R0010	Establish a training priority	Yes	002.03
R0310	Maintain the departmental Watch and Station Bill	Yes	002.03
R0330	Monitor the training progress of personnel	Yes	002.03
R0730	Plan Combat Systems (CS) collective/team training (e.g. ERT/HAZMAT exercises)	Yes	002.03
R0750	Produce a departmental annual training plan	Yes	002.03

Task No	Task Statement	Train	PO/EO
R1040	Determine subordinate's individual training and educational requirement	Yes	002.03
R1140	Manage W ENG department personnel readiness	Yes	002.02
S1310	Conduct receipt inspection of MK 48 Torpedo	Yes	004.02
S1320	Document Maintenance IAW SUBSAR regulations	Yes	003.02
S1330	Monitor CSED support IAW Escape and rescue certificate	Yes	003.02
S1340	Monitor CSED Support for Towed Array Operations (TAO)	Yes	003.04
S1500	Analyze/evaluate/interpret a Tactical Television System functional check result	Yes	003.02
S1510	Direct a corrective maintenance/repair activity on a Tactical Television System	Yes	003.02
S1590	Analyze/evaluate/interpret a Desiccant System functional check result	Yes	003.02
S1600	Direct a corrective maintenance/repair activity on a Desiccant System	Yes	003.02
S1610	Support towed array operations	Yes	003.04
S1620	Supporting Weapons Handling & Discharge System (WHDS) training	Yes	004.02
S1630	Supporting weapons employment for Weapons Certification Programme (WCP)	Yes	004.03
S1640	Inspect MK48 torpedo/Torpedo Mounted Dispenser (TMD)	Yes	004.02
S1650	Conduct duties of CS Subject Matter Expert (SME) in HQ 1	Yes	PO 005
S1660	Conduct the duties of small arms custodian	Yes	003.01
S1670	Managing Submerged Signal Ejector (SSE) stores	Yes	006.05
S1680	Monitor a Tactical Television System	Yes	006.03
S1690	Direct a Tactical Television System functional check result	Yes	003.02
S1700	Direct a Tactical Television System preventive maintenance	Yes	003.02
S1710	Analyze a Tactical Television System test/trial/inspection result or report	Yes	003.02
S1720	Monitor a Desiccant System	Yes	006.03
S1730	Direct a Desiccant System functional check result	Yes	003.02
S1740	Direct a Desiccant System preventive maintenance	Yes	003.02
S1750	Analyze a Desiccant System test/trial/inspection result or report	Yes	003.02

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 18 Jan – 05 Feb 2016

Location: CFNOS S-17, Sub Div Chris Saunders Conference Room

Board Members:

Position	Surname	Initials	Rank	MOSID	Unit
Chair	Moore	MA	Lt(N)	00344	CFNOS Sub Div
TDO	Surette	RA	Capt	00211	CFNES QA
Secretary	Tanguay	GC	MS	00366-06	HMCS Corner Brook
Trg Mgr	Froment	D	CPO2	00299	NPTG HQ
Stds Rep	Bartlett	J	PO1	00366-08	CFNOS Stds
Coastal Occ Advisor	Sevigny	LAD	CPO2	00366-08	Marlant HQ CCFL
SME	Larrivee	P	LCdr	00344	ST(A)
SME	Perry	A	PO1	00366-08	CFNOS Sub Div
SME	Langlois	E	PO1	00366-08	CFNOS Sub Div
SME	Anthony	JD	PO2	00366-05	CFNOS Sub Div

Board Objective:

1. Review/confirm W Eng MTL pertaining to AILN;
2. Redesign AILN to become an RQ-PO1, aligning course PO's with those of AKPL;
3. Update Submarine systems/components to reflect those currently fitted;
4. Review/update HAZMAT trg requirements;
5. Recommend opportunities for Technology-Enabled Learning (TEL); and
6. Update/Align QSP with NPTG QSP template IAW NPTORD 4500-22.

Specification Task – No Train Decisions

*Note: Due to the MTL not being complete at the time of the QSPWB, the QSPWB reviewed all tasks to ensure everything for this job was captured for job/training purposes. Tasks determined to be N/A to this job are not included in this No Train list.

Task No	Task Statement	Substantiation
A0010	Draft an Unsatisfactory Condition Report (UCR)	Previously trained
A0020	Draft an Engineering Change (EC) Proposal	Previously trained
A0030	Investigate an EC Proposal	Previously trained
A0050	Forecast a budget expenditure	Previously trained
A0060	Implement a change to a work procedure	Gained through previous experience
A0070	Liaise with a Field Service Representative (FSR)	Gained through previous experience
A0080	Action a TRAN REQ	Gained through previous experience
A0100	Draft a Work Order or notification in DRIMS	Previously trained
A0110	Draft a maintenance Work Order in DRIMS	Previously trained
A0130	Draft a Pre-Installation Failure Report (PIF)	Previously trained
A0140	Draft a Spares List for deployed operations	Gained through previous experience
A0150	Draft a Statement of Work (SOW)	Gained through previous experience
A0160	Draft a trial agenda	Gained through previous experience
A0200	Edit/re-write a trial agenda (e.g. red line)	Gained through previous experience
A0220	Exercise Due Diligence WRT contractor activities	Gained through previous experience
A0260	Identify an FMF production problem	Gained through previous experience
A0310	Investigate a maintenance Work Order	Gained through previous experience
A0320	Investigate a DRIMS Work Order	Gained through previous experience
A0330	Investigate a PIF	Gained through previous experience
A0340	Investigate a SOW	Gained through previous experience
A0360	Investigate an OPDEF	Gained through previous experience
A0370	Investigate an UCR	Gained through previous experience
A0380	Liaise with ORO WRT CS maintenance activities	Gained through previous experience
A0390	Liaise with other sections/units on work related matters	Gained through previous experience
A0410	Maintain Maintenance records in DRMIS	Previously trained on DRMIS courses
A0420	Maintain the Defence Resources Information Management System DRIMS	Previously trained on DRMIS courses
A0440	Manage an operating budget	Previously trained via DL Expenditure Management
A0460	Manage quality control over workmanship and materials	Gained through previous experience
A0480	Manage work period work request submissions	Gained through previous experience
A0500	Originate a demand for stores	Previously trained on 118936
A0510	Perform a local purchase for stores	Previously trained on 118936
A0520	Perform CM or PM in DRMIS	Previously trained on DRMIS courses
A0540	Approve a DRIMS maintenance Work Order or Notification	Previously trained on DRMIS courses
A0570	Approve a demand for stores	Gained through previous experience
A0590	Approve another unit Work Order in DRIMS	Previously trained on DRMIS courses
A0630	Provide input into a business plan (e.g. Capability plan)	Gained through previous experience
A0650	Provide input to Post-Deployment Report	Gained through previous experience
A0750	Review a PM schedule	Gained through previous experience

A0760	Review a Spares List for deployed operations	Gained through previous experience
A0770	Review a trial agenda	Gained through previous experience
A0780	Assign resources to accomplish a task, job or project	Gained through previous experience
A0790	Review Unit Post-Deployment reports	Gained through previous experience
A0810	Track a demand for stores	Gained through previous experience
A0850	Validate the Fitted Equipment List (FEL)	Gained through previous experience
A0890	Complete Personnel Aloft procedures/requirements	Previously trained on the Maritime Command RADHAZ Officer Course (115868), which the QSPWB determined is now a course prerequisite to this course.
A0900	Conduct a ship-level test/trial/inspection	Previously trained on Sub OCC
A0910	Conduct CS suite level test/trial/inspection (e.g. CSAT)	Previously trained on Sub OCC
A1020	Coordinate Personnel Aloft requirements	Previously trained on the Maritime Command RADHAZ Officer Course (115868), which the QSPWB determined is now a course prerequisite to this course.
A1030	Coordinate the Formation response to an OPDEF	Gained through previous experience
A1050	Coordinate the installation or removal of equipment	Gained through previous experience
A1060	Determine the resources (labour, material, time and documentation) required to accomplish a task, job or project	Trained on specialty courses
A1100	Direct a CS suite level test/trial/inspection	Gained through previous experience
A1120	Direct a ship-level test/trial/inspection	Gained through previous experience
A1130	Direct a tactical system integration test/trial/inspection	Previously trained on Sub OCC
A1140	Direct a Weapon system integration test/trial/inspection	Previously trained on Sub OCC
A1150	Direct an acoustic system integration test/trial/inspection	Previously trained on Sub OCC
B0050	Draft a Significant Incident Report	Gained through previous experience
B0060	Investigate/review a Significant Incident Report	Gained through previous experience
B0070	Direct TEMMIS activities	Gained through previous experience
B0080	Direct the handling of ammunition/explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0090	Dispose of ammunition/explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0110	Draft a Monthly Ammunition Expenditure Report	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0130	Draft an ammunition defect and malfunction report (CF 410)	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.

B0140	Draft an Ammunition Order	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0150	Draft an Ammunition Supply Document	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0160	Embark/disembark ammunition and explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0170	Enforce ammunition compatibility criteria	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0190	Enforce Occupational Health and Safety Regulations (Canada Labour Code)	subsequent training on Technical Administration
B0200	Identify ammunition types	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0210	Implement an Ammunition Directive	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0220	Inspect ammunition and explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0240	Investigate a CF 410	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0260	Issue ammunition for use	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0280	Maintain ammunition data cards (CF 1040)	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0290	Maintain an ammunition log	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0300	Maintain classified documents	Gained through previous experience
B0340	Monitor reports and returns regarding explosives, ammunition, ranges and procedures	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0360	Perform magazines and lockers preventative and corrective maintenance	AKQN
B0370	Perform preventative and corrective maintenance on ammunition lifting devices	AKQN
B0380	Perform preventive maintenance of ammunition/explosives	AKQN
B0450	Perform the duties of an Ammunition Officer	Previously trained on AIRX (not required for CSEC)
B0490	Perform the duties of the Departmental HAZMAT Representative	Gained through previous experience
B0520	Perform the duties of the Formation/Unit Non-Ionizing Electromagnetic Radiation Hazards (RADHAZ) Officer	Previously trained on the Maritime Command RADHAZ Officer Course (115868), which the

		QSPWB determined is now a course prerequisite to this course.
B0540	Perform the duties of the General Safety Officer	Previously trained on formation UGSO Course
B0560	Approve a Monthly Ammunition Expenditure Report	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0580	Provide advice on the ammunition safety program	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0590	Approve a waiver for shipboard burning and welding in the vicinity of an ammunition space or launcher	N/A to this job
B0600	Approve an Ammunition Supply Document	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0610	Return defective ammunition	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0620	Action an Ammunition Restriction	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0630	Review a Monthly Ammunition Expenditure Report	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0640	Review an Ammunition Order	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0650	Review an Ammunition Supply Document	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0660	Rotate ammunition	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0670	Stow ammunition and explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0680	Supervise the embarking/disembarking of ammunition and explosives	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0690	Calculate net explosive quantity	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0700	Certify an ammunition container Free From Explosive	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0710	Certify an ammunition or explosive stowage space following a refit or new construction	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0760	Conduct a Safety Inspection	Gained through previous experience

B0780	Conduct annual ammunition inspections of magazines and storage facilities	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0790	Conduct Equipment Health Monitoring	Gained through previous experience
B0810	Conduct magazine rounds	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0820	Conduct non-ionizing radiation safety measurements	Previously trained on the Maritime Command RADHAZ Officer Course (115868), which the QSPWB determined is now a course prerequisite to this course.
B0830	Conduct pre-acceptance inspection of ammunition/explosive	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0840	Coordinate the ammunition safety program	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0850	Advise on the design or modification of an ammunition storage facility	Previously trained on the Ammunition Custodian Course (118948), which the QSPWB determined is now a course prerequisite to this course.
B0870	Direct a HAZMAT cleanup team	Redundant to B0890
B0880	Direct Equipment Health Monitoring	Gained through previous experience
F0070	Initialize a Submarine Fire Control System (SFCS) to an operational state	Previously trained on AIMZ
F0300	Operate a Submarine Periscope System	Previously trained on ALDE
F0370	Perform a SFCS functional check	Previously trained on AIMZ
F0390	Perform a Submarine Periscope System functional check	Previously trained on ALDE
F0570	Perform the duties of the Left Console Operator	Previously trained on AIMZ
F0580	Perform the duties of the Right Console Operator	Previously trained on AIMZ
F0590	Perform the duties of the Safety Observer during the embarking/disembarking of a torpedo	Redundant to F0740
F1030	Configure/reconfigure SFCS software	Previously trained on AKQN
F1160	Analyze/evaluate/interpret a SFCS test/trial/inspection result/report result	Redundant to F1150
G0580	Perform the duties of the acoustic sensor handling system Safety Observer	Previously trained on AIDI
H0090	Initialize an Underwater Telephone (UWT) System to an operational state	Previously trained on AILS Part 1
H0240	Operate a SSE System	Previously trained on AILS
H0260	Operate a Bathy System	Previously trained on AILS
H0270	Operate a Torpedo Decoy System	Previously trained on AILS
H0280	Operate an UWT System	Previously trained on AILS Part 1
H0500	Perform the duties of the Safety Observer	Gained through previous experience
I0080	Install a new or repaired/refurbished/overhauled Internal Communication System component	Previously trained on AKQK

I0190	Operate a Submarine Escape Communication System	Previously trained on AILS Part 1
I0200	Operate an Internal Communication System	Previously trained on AILS Part 1
I0290	Perform a Submarine Escape Communication System functional check	Previously trained on AKQK
I0690	Configure a Submarine Escape Communication System to meet an operational requirement	Previously trained on AKQK
J0280	Operate a VHF System	Previously trained on AILS
K0410	Raise/lower an antenna	Previously trained on AILS
M0190	Operate a GPS system	Previously trained on ALDE
M0200	Operate an AIS System	Previously trained on ALDE
M0210	Operate a NAV Radar system	Previously trained on ALDE
M0230	Operate an Internal Navigation System	Previously trained on ALDE
M0850	Configure a NAV Radar system to meet an operational requirement	Previously trained on ALDE
N0010	Initialize a Depth Sounder System to an operational state	Previously trained on ALDE
N0210	Operate a Depth Sounder System	Previously trained on ALDE
N0250	Operate a SHINADS system	Previously trained on ALDE
N0820	Configure a SHINADS system to meet an operational requirement	Previously trained on ALDE
O0660	Analyze/evaluate/interpret a Link System functional check result	Redundant O0630
O0670	Analyze/evaluate/interpret a Link System test/trial/inspection result or report	Redundant O0640
O0900	Direct a corrective maintenance/repair activity on a Link System	Redundant with task O0870
O0930	Direct a Link System functional check	Redundant O0910
O0940	Direct a Link System PM activity (including a test/trial/inspection)	Redundant with task O0920
Q0180	Monitor an Emergency Beacon performance	Redundant with task I0150
Q0220	Operate an Emergency Beacon	Previously trained on AILS
Q0910	Direct a corrective maintenance/repair activity on an Emergency Beacon	Redundant with task I1170
Q0980	Direct an Emergency Beacon functional check	Redundant with task I1160
Q0990	Direct an Emergency Beacon PM activity (including a test/trial/inspection)	Redundant with task I1180
R0020	Draft a lesson plan	Previously trained on PLQ
R0030	Monitor an instructor	Trained at CFTDC (IS course)
R0040	Draft an Establishment Change Proposal (ECP)	Gained through previous experience
R0050	Review an ECP	Gained through previous experience
R0080	Monitor the occupation Preferred Manning Level (PML) (e.g. recruitment, promotion, attrition)	Gained through previous experience
R0090	Post or attach post personnel	Gained through previous experience
R0100	Perform the duties of an Assisting Officer in support of a Redress of Grievance	Taught via DLN Assisting Officer course
R0110	Monitor training throughput	Gained through previous experience

R0120	Draft a course report (CF 377)	Gained through previous experience
R0130	Draft a Personnel Evaluation Report (PER) or a Personnel Development Review (PDR)	Previously trained on PLQ
R0150	Draft a Training Deficiency (Trg Def)	Gained through previous experience
R0160	Enforce the application of the Naval Divisional System	Gained through previous experience
R0170	Escape/Rescue from disabled submarine	Previously trained on AILS
R0180	Evaluate a course critique	Gained through previous experience
R0190	Forecast a training requirement	Gained through previous experience
R0210	Instruct personnel (e.g. formal coursing, OJT etc.)	Previously trained on PLQ
R0220	Investigate a formal or informal complaint (e.g. harassment, disciplinary)	Previously trained on ILP
R0230	Liaise with other government agencies	Previously trained on ILP
R0250	Invigilate a PC or an EC	Gained through previous experience
R0260	Liaise with emergency services agencies	Gained through previous experience
R0280	Maintain Divisional notes	Gained through previous experience
R0290	Maintain personal NBCD protective equipment	Gained through previous experience
R0320	Monitor QSP compliance	Previously trained on IS course
R0340	Operate a fitted fire suppression system	Previously trained on AILS
R0350	Organize a summary trial	Previously trained on ILP
R0370	Perform adjacent compartment rounds	Previously trained on AILS
R0380	Perform an immediate action drill for NBCD	NIAC
R0400	Perform as a member of a damage control party	Previously trained on AILS
R0410	Perform as a member of a force protection organization or a base security force	Previously trained on AILS
R0420	Perform as a member of the base duty watch organization	Gained through previous experience
R0430	Perform duties of a member of a ship firefighting team	Previously trained on AILS
R0510	Perform the duties of a Magazine Yeoman in foreign port	Previously trained on DWS / OOD
R0590	Perform the duties of the MCC/DCC or SCC/MCC Watchkeeper	Previously trained on AILS
R0600	Perform the duties of a member of the mess committee	Gained through previous experience
R0610	Perform the duties of a member of the ship's committee (e.g. ship welfare, entertainment, Sports O etc.)	Gained through previous experience
R0620	Perform the duties of a Petty Officer of the Watch	Previously trained on AILR
R0630	Perform the duties of a Senior Hand of a mess	Gained through previous experience
R0640	Perform the duties of an Assisting Officer in support of a summary trial	Previously trained on DLN
R0660	Perform the duties of the duty PMC	Gained through previous experience
R0670	Perform the duties of the President of the Mess Committee (PMC)	Gained through previous experience

R0680	Perform the duties of the Attack Team Leader	Previously trained on AILS
R0690	Perform the duties of the Duty Technician (all classes)	Previously trained on AILS
R0780	Provide input to the unit/section ranking of personnel	Gained through previous experience
R0790	Recommend a disciplinary or an administrative measure in response to a complaint	Previously trained on ILP
R0800	Recommend administrative sanctions for a subordinate	Previously trained on ILP
R0820	Assign an instructor	Gained through previous experience
R0830	Resolve a formal or informal complaint	Previously trained on ADR
R0840	Respond to a survival drill	Previously trained on AILS
R0850	Review a course report (CF 377)	Gained through previous experience
R0860	Assign instructional resource and material (e.g. classroom/equipment/training areas)	Gained through previous experience
R0870	Review a lesson plan	Gained through previous experience
R0880	Review a PER or a PDR	Previously trained on PLQ
R0890	Review a Trg Def	Gained through previous experience
R0900	Secure for sea	Gained through previous experience
R0910	Select personnel for training/coursing	Gained through previous experience
R0920	Assist in the conduct of a summary trial	Previously trained on ILP & Presiding Officer Course
R0930	Store ship	Gained through previous experience
R0940	Supervise the cleaning of a ship's compartment or passageway	Gained through previous experience
R0950	Organise/prepare the conduct of a summary trial	Previously trained on ILP & Presiding Officer Course
R0960	Track collective and individual training	Gained through previous experience
R0970	Brief and debrief personnel with regards to PER or PDR results	Previously trained on PLQ
R0990	Clean a ship's compartment or passageway	Gained through previous experience
R1000	Conduct a Training Review Board (TRB)	Gained through previous experience
R1010	Administer a course critique	Gained through previous experience
R1020	Conduct a shipboard bomb search	Gained through previous experience
R1030	Coordinate training facilities usage with contractors	Gained through previous experience
R1050	Develop a Performance Check (PC) or an Enabling Check (EC)	Previously trained on PLQ
R1060	Develop a Qualification Standard Plan (QSP) (board member)	Gained through previous experience
R1120	Maintain Monitor Mass departmental personnel data	Previously trained on Monitor MASS course
R1130	Maintain Monitor Mass departmental training data	Previously trained on Monitor MASS course
S1140	Perform the Duties of 35 BLKD watchkeeper	Previously trained on AKYJ
S1150	Operate all Submarine Hydraulic systems	Previously trained on AILS part 2
S1160	Operate all Submarine HP/LP air systems	Previously trained on AILS part 2
S1170	Operate Trim Bilge and Ballast systems	Previously trained on AILS part 2
S1180	Operate Submarine domestic systems	Previously trained on AILS part 2

S1190	Operate Submarine cooling systems	Previously trained on AILS part 2
S1200	Operate Submarine Electrical systems	Previously trained on AILS part 2
S1210	Operate Submarine OMC	Previously trained on AILS part 2
S1220	Navigate Submarine on the Surface	Previously trained on AILR
S1230	Supervise Control room on the Surface	Previously trained on AILR
S1240	Perform the duties of SCOOW on the Surface	Previously trained on AILR
S1250	Direct and Manage control room Dive	Previously trained on ALDE
S1260	Navigate Submarine Dive	Previously trained on ALDE
S1270	Perform the duties of periscope watch keeper	Previously trained on ALDE
S1280	Perform the duties of SCOOW dive	Previously trained on ALDE
S1290	Operate Weapons Control Console WCC	Previously trained on AIMZ
S1300	Perform the duties of HQ1 SME during EOP	Redundant with T0018
S1350	Plan logistic support for TAO in foreign Countries	Previously trained on AIDI
S1360	Conduct test and trials for Escape equipment	Previously trained on AILF
S1370	Conduct Pre and Post Escape inspection for Submarines	Previously trained on AILF
S1380	Conduct Escape and Rescue Training	Previously trained on AILF
S1390	Coordinate load out of SUBSAR reaction equipment	Previously trained on AILF
S1400	Conduct Training Facility inspections	Previously trained on AILF
S1410	Review Test and trials report for escape equipment	Previously trained on AILF
S1430	Initialize Tactical Television System to an operational state	Gained through previous experience

Discussions / Decisions:

Monday 18 Jan 2016
As per 121649Z JAN 16, the Qualification Standard & Plan Writing Board (QSPWB) for the Submarine Combat System Engineering Coordinator/Combat System Engineering Officer course was convened @0830hrs.

Following a Welcome Brief by the QSPWB Chair, and a further presentation outlining the general format and conduct of a QSPWB by the TDO; all members of the QSPWB reviewed the MTL (provided by CPO1 Mills D NAV P&T, Occ Manager) which was found to be lacking in several aspects of Submarine specific tasks. As such, the QSPWB began a comprehensive review of all tasks within the MTL pertaining to the job of a CSEC/O, determining which tasks were trained or not trained on the job.

Tuesday 19 January 2016
The QSPWB began each day with a recap of the progress thus far.

The QSPWB then continued with the MTL review as well as reviewing the existing task list in the current CSEC/O QSP. The MTL review was completed ~1030hrs, at which time the Secretary and TDO compiled the trained Task List and itemized them for Scalaring purposes.

As stated in the convening order, one of the boards mandate is to “REDESIGN AILN TO BECOME AN RQ-PO1, ALIGNING COURSE PERFORMANCE OBJECTIVES WITH THOSE OF AKPL”. As such, the titles for each PO from the AKPL course provided the basis for the POs herein, from which each task from the task list was grouped within these POs. Necessary MTL task additions were discussed at length, and added with QSPWB concurrence.

Course prerequisites were mentioned (namely Unit RADHAZO course as well as the Magazine Custodian course). At this point, no decision was made, but a general consensus has begun to emerge in which the two mentioned courses will be stated as pre-requisites within the QSP. It should be noted that the Magazine Custodian course is currently is a prerequisite for the AKPL qualification; making this course a prerequisite for the CSEC/O course would further align the two qualifications.

Wednesday 20 January 2016
Following the daily SITREP, QSPWB members then began an open discussion regarding the necessity of the Unit RADHAZ O qualification as it pertains to the duties of the CSEC/O. Group consensus was that all potential trainees on the Submarine CSEC/O course should hold this qualification prior to commencing the Submarine CSEC/O course; as such, the Unit RADHAZ O course was deemed an official pre-requisite to the AILN course.

Following this, the board began compiling and organizing the various submarine related tasks from the MTL into POs/EOs. Scalaring was complete by COB.

Thursday 21 January 2016
Following the daily SITREP, the QSPWB reviewed the various POs/EOs as they were organized the previous day, as well as drafted the QSP Ch. 1. Following the PO/EO review & concurrence, the QSPWB reviewed the draft of Ch. 1. Several items and aspects were discussed. It was in these discussions, that the Assessor role in Chapter 1 was removed. It was decided that the Incident Controller requirement is not necessary. Additionally, the Support Manning for the HQ1 phase was removed as a requirement because an effective assessment of a trainee could be conducted without such support manning. It was added as recommended staff support in Chapter 1, as having this support would improve efficiency during the evaluations of the trainees.

Due to the manning of positions in the trainers/simulator, optimum loading was decided to be 3 vice 4 trainees. It was also decided that due to present personnel issues within the Submarine Fleet, the minimum load was reduced from 2 to 1 trainee.

Friday 22 Jan 2016
Following the daily SITREP, the QSPWB began a thorough re-review of Chapter 1 & 2 drafts, with only minor amendments required. Next, a PO/MTL cross-referencing review began, which resulted in additional tasks were added to the MTL.

Maintenance Process. All steps in the maintenance process have been included for equipment not previously included in the MTL, to include Monitor, Direct CM, Direct PM,

Direct Functional Check, Analyze test/trial. Tasks to cover all steps in the maintenance process have not been added to the MTL for each piece of equipment existing in the MTL, as it is outside the scope of this QSPWB, however it is understood that the maintenance process in full applies to each piece of equipment pertaining to the job of the CSEC/O.

The reviewed versions of Ch1 and Ch2 received all QSPWB members' concurrence. These drafts, as well as the ROP and the revised MTL containing an extra tab listing tasks missing from the original MTL that were added, with justification, to better standardize the tasks within the MTL) were sent via the Trg Mgr to NPTG for provisional approval and the Occ Mgr as a courtesy prior to EO development.

Monday 25 January 2016
Following the daily SITREP, the TDO gave a brief presentation on the development and drafting of EOs. At which time, the QSPWB collectively began drafting all of the EOs contained within PO 004. Following this, the QSPWB drafted EO 005.01, which it was decided that the entire PO could be contained within a single EO, much to the chagrin of the TDO.

Tuesday 26 January 2016
Following the daily SITREP, the QSPWB began a thorough review of EO 005.01 and in particular, the pertinent references for each teaching point within the EO.
Following this review, the QSPWB created EO 001.01 (HAZMAT). Currently, the Submarine HAZMAT program is under review by Sea Training (Submarines), and it is noted that this PO will likely require amendments once this comprehensive review is completed. Nevertheless, the QSPWB carried on.

It was during the group discussion that task number B0310 (Maintain equipment Associated with HAZMAT clean up) was a "No Train" task due to the fact that the CSEC/O would never be called upon to Maintain the equipment.
The remainder of the day was spent drafting all of the supporting EOs for POs 001, 003, 004, and 005.

Wednesday 27 January 2016
Following the daily SITREP, the coastal MOS advisor informed the QSPWB that the MOS Manager had raised some concerns WRT the RADHAZ O qualification being a prerequisite for this course due to the fact that the QL6B for the Surface fleet doesn't have such a prerequisite and the course is actually incorporated into the QL7 course vice the QL6B. This constitutes an imbalance between the Submarine qualification and that of the General Service. The board discussed the issue and agreed to revisit prior to concluding the QSPWB.

The QSPWB developed EO 006.01 and decided on a template for the remaining EOs within PO 006. Following this, the board Secretary continued the drafting of the remaining EO's within PO 006 using EO 006.01 as a guide. While he continued with the EOs for PO 006, the remainder of the board discussed and drafted both EOs 002.01 and 002.02 (within which, it was decided that the EO 002.03 was too small to warrant a full EO and that it could be encapsulated within EO 002.02).

By COB, all of the initial drafts of the remaining EOs were completed.

Thursday 28 January 2016
Following the daily SITREP, the Chair provided the board with the daily benchmarks and goals. Following this, the TDO gave a presentation on assessments and EC/PC development. The QSPWB then concentrated their efforts on drafting the ECs and PC for PO 002.

Friday 29 January 2016
Following the daily SITREP, the QSPWB decided to complete all of the ECs for 001 and 006 prior to completing the remaining PCs. The board adjourned for the day with only the PCs for 003 and 006 left to be completed.

Monday 1 February 2016
Following the daily SITREP, the board drafted of the PC 003 and 006. Once complete, the board reviewed the QSP draft in its entirety for clarity and clerical errors, as well as discussed which EOs should be recommended for TEL. Several were identified, and included in summary form as Annex N in this QSP. Following this initial review, the QSPWB addressed outstanding “Parking Lot” items such as the prerequisites (RADHAZO course, and the Magazine Custodian) as well as the inclusion of a Project Management EO or TP. The QSPWB took a preliminary vote and the board was split between whether the RADHAZ O course is included within the CSEC/O course or is listed as a pre-requisite. The Board Chairperson decided to hold over the topic until the following day at which time the coastal W Eng MOC Advisor would be able to weigh in and provide a MOS point of view.

Tuesday 2 February 2016
Following the daily SITREP, a comprehensive debate WRT course prerequisites, it was decided that both these qualifications will be designated as pre-requisites. The rationale for the Magazine Custodian course as a prerequisite is that it more closely aligns the Submarine CSEC/O job with the general service W Eng QL6B course (NQual AKPL). The RADHAZO course was also designated as a prerequisite, as the trainees will be required to conduct the duties of RADHAZO when they are posted to a Submarine as a CSEC. The RADHAZO course has already been included within the W Eng QL7 course but it was the QSPWB’s opinion that both the CSEC/O and the QL 7 courses are designed to prepare a trainee to act as the Senior W Eng onboard an operational unit. Therefore, for Submariners there is a requirement to have this qualification earlier in their career.

The Board concluded its collective activities, and allocated post-board activities/ documentation to the various SMEs. A sole issue remained unaddressed, whether a Project Management EO should be included within the CSEC/O course. It was the QSPWB’s collective opinion that given the duties and responsibilities expected of a CSEC (NCS Eng Officers receive Project Management training during the Naval Engineering Common course), and in the spirit of aligning as closely as possible with the W Eng QL6B (which includes a one day course offered via the LCC), a Project Management EO should be included, with a duration of 1-1.5 trg days, IOT provide an

overview of the Project Management process and the various aspects which will have a direct effect on Planned Maintenance periods. Using the EO resident within the Naval Engineering Common QSP as a template, curtailed to what would be included within the LCC offered course.

Tuesday 2 February 2016
QSPWB concurred that the mandate of the QSPWB have been met, and the QSP meets the trg requirements of the job of the CSEC/O.

Post-QSPWB, the QSP was QA'd by CFNES staff and sent out for final concurrence to all QSPWB members. All concurred, with only minor changed required.

Lessons Learned: Email sent to D Nav P&T requesting a more thorough review/confirmation of MTL prior to convening a QSPWB. This MTL was severely lacking in tasks performed on the job and a significant amount of time was spent updating it WRT the tasks performed by the CSEC/O.

Guidance to Instructional Staff: Follow Remarks within each EO for specific trg details.

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	LSRs for TEL recommendations	PO1 Perry	Prior to submission of QSP for approval
2			

Other Considerations (if applicable): Nil

ANNEX D – QSPWB SCALAR

PO Title	EO Title(s)
PO 001- Coordinate Unit HAZMAT Responsibilities	EO 001.01 - Manage Unit HAZMAT Programme Readiness
	EO 001.02 - Direct Response to a HAZMAT Incident
PO 002 - Maintain Divisional Administration Readiness	EO 002.01 - Manage CS Department Safety IAW SEMS
	EO 002.02 - Manage CS Department Personnel Readiness
PO 003 - Coordinate Divisional Technical Readiness	EO 003.01 - Manage Technical Administration
	EO 003.02 - Coordinate Ship-Level Maintenance Activities
	EO 003.03 - Manage CS Department Pre-Sail Checks
	EO 003.04 - Support Towed Array Operations
	EO 003.05 - Contribute to the Project Management Process
PO 004 - Support Weapons Certification Programme	EO 004.01 - Support WCP Administration
	EO 004.02 - Support WHDST
	EO 004.03 - Support Weapons Employment for WCP
PO 005 - Conduct the Duties of CS Engineering SME in HQ1	EO 005.01 - Sustain Combat Systems in Emergency Situations
PO 006 - Advise Command on the CS Suite	EO 006.01 - Advise Command on the Communication Suite
	EO 006.02 - Advise Command on the Electronic Warfare Suite
	EO 006.03 - Advise Command on the Navigation Suite
	EO 006.04 - Advise Command on the Sonar Suite
	EO 006.05 - Advise Command on the Fire Control Suite
	EO 006.06 - Advise Command on CS Suite Integration

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

N/A

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
AIS	Automatic Identification System
AGA	Air-Ground-Air
ATP	Air Turbine Pump
ATT	Actual Training Time
BAT	Base Administrative Time
BSQ	Basic Submarine Qualification
CCA	Curriculum Control Authority
CCSF	Commander Canadian Submarine Force
CF	Canadian Forces
CFCD	Canadian Forces Controlled Document
CFITES	Canadian Forces Individual Training Education System
CFNOS	Canadian Forces Naval Operations School
CHOD	Chief-of-Department
CMP	Chief of Military Personnel
CNPTG	Commander Naval Personnel and Training Group
CM	Corrective Maintenance
CO	Commanding Officer
COS	Change-Over Switch
COTM	Commanding Officer Temporary Memorandum
CRR	Combat Readiness Requirements
CRWK(D)	Control Room Watchkeeper (Dived)
CRWK(S)	Control Room Watchkeeper (Supervisor)
CS	Combat Systems
CSE	Combat Systems Engineering
CSEC	Combat Systems Engineering Coordinator
CSED	Combat Systems Engineering Department
CSEC/O	Combat Systems Engineering Coordinator/Officer
DAOD	Defence Administrative Orders and Directives
DC	Damage Control
DL	Distance Learning
DND	Department of National Defence
DRMIS	Defence Resource Management Information System
DWAN	Defence Wide Area Network

Acronym and Abbreviation	Term or Definition
EC	Enabling Check
EC	Engineering Change
EO	Enabling Objective
ERT	Emergency Response Team
EUWT	Emergency Underwater Telephone
EW	Electronic Warfare
FC	Fire Control
FLAD	First-Level Area Diagram
FNTS	Future Naval Training Strategy
FTA	Fleet Technical Authority
FY	Fiscal Year
GCCS-M	Global Command & Control System Maritime
HAT	Harbour Acceptance Trial
HOD	Head of Department
HQ1	Headquarters 1
MARCOM	Maritime Command
HF	High Frequency
HMC	Her Majesty's Canadian
HAZMAT	Hazardous Material
IAW	In Accordance With
IC	Incident Controller
IFF	Interrogator Friend-or-Foe
ISDS	Information Screening and Delivery System
LAN	Local Access Network
LF	Low Frequency
LFC	Latent Failure Checks
LOTO	Lockout / Tagout
MATCERT	Material Certification
MCC	Machinery Control Console
MF	Medium Frequency
MITE	Military Individual Training and Education
Mk	Mark
MRL	Maritime Rear Link
MS	Master Seaman

Acronym and Abbreviation	Term or Definition
min	Minute
MTTL	Minimum Tube Total Loading
NaMMS	Naval Material Management System
NAVORD	Naval Order
NCM	Non-Commissioned Member
NCS ENG	Naval Combat Systems Engineering
NDDS	Navigation Data Distribution System
NDSOD	National Defence Security Orders & Directives
NEM	Naval Engineering Manual
NIOSH	National Institute for Occupational Safety and Health
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NTS	Naval Training System
NQual	National Qualification
OBT	On-Board Training
OFM	Otto Fuel Monitor
OJPR	On-Job Performance Report
OPDEF	Operational Deficiency
OQE	Objective Quality Evidence
PC	Performance Check
PEL	Periodic Engineering Letter
PLAR	Prior Learning Assessment and Recognition
PM	Preventive Maintenance
PO	Performance Objective
PO2	Petty Officer 2nd Class
PRB	Progress Review Board
PT	Physical Fitness
QSP	Qualification Standard and Plan
QSPWB	Qualification Standard and Plan Writing Board
RADHAZO	Radiation Hazard Officer
RCN	Royal Canadian Navy
Ref	Reference
RQ	Rank Qualification
SAT	Satisfactory

Acronym and Abbreviation	Term or Definition
SAT	Sea Acceptance Trial
SCC	Ship Control Console
SDM	SHINNADS Dual Monitor
SEMS	Safety & Environment Management System
SFCS	Submarine Fire Control System
SHINNADS	Shipboard Integrated Navigation And Display System
SME	Subject Matter Expert
SMORO	Submarine Operations Room Officer
SMTI	Submarine Temporary Instruction
SOP	Standard Operating Procedures
SSE	Submerged Signal Ejector
STAHT	Submarine Towed Array Handling Team
ST(S)	Sea Training (Submarines)
STW	Set-to-Work
Sub IP	Submarine Internet Protocol
SUBSAFE	Submarine Safety
SUBSAR	Submarine Search & Rescue
SUBWOPS	Submarine Weapons Operating Procedures
TA	Training Authority
TAT	Training Administrative Time
TBD	To Be Determined
TE	Training Establishment
TDC	Training Development Centre
TMD	Torpedo Mounted Dispenser
TP	Teaching Point
TTT	Training Travel Time
TRB	Training Review Board
TRP	Tiered Readiness Program
Trg	Training
TWS	Tactical Weapons Systems
UCR	Unsatisfactory Condition Report
UHF	Ultra High Frequency
UNSAT	Unsatisfactory
UWT	Underwater Telephone

Acronym and Abbreviation	Term or Definition
VCEOP	Victoria Class Emergency Operating Procedures
VCS	Victoria-class Communications System
VCSOP	Victoria Class Standard Operating Procedures
VCSSO	Victoria Class Ship's Standing Orders
VICES	Voice Internal Communications Equipment Submarine
VHF	Very High Frequency
VLF	Very Low Frequency
VCSWG	Victoria Class Submarine Work-ups Guide
VSTI	Victoria Submarine Technical Instruction
WAN	Wide Access Network
WCP	Weapon Certification Programme
W Eng	Weapons Engineering
WHDS	Weapons Handling and Discharge System
WHDST	Weapons Handling and Discharge System Trainer
WHMIS	Workplace Hazardous Materials Information System
WSC	Weapon Stowage Compartment
WSDB	Weapon System Data Bus
WUPS	Work-ups

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO / EO	PO / EO Title	EO	EC*	PC*	PO
PO 002	**Maintain Division Admin Readiness**	**12**	**2**	**3**	**17**
EO 002.01	Manage CS Department Safety IAW SEMS	7			
EC 002.01	Manage CS Department Safety IAW SEMS		1		
EO 002.02	Manage CS Department Personnel Readiness	5			
EC 002.02	Manage CS Department Personnel Readiness		1		
PC 002	Maintain Division Admin Readiness			3	
PO 003	**Coordinate Divisional Technical Readiness**	**48**	**5**	**1**	**54**
EO 003.01	Manage Technical Administration	19			
EC 003.01	Manage Technical Administration		1		
EO 003.02	Coordinate Ship-Level Maintenance Activities	15			
EC 003.02	Coordinate Ship-Level Maintenance Activities		1 (review / debrief)		
EO 003.03	Manage CS Department Pre-Sail Checks	2			
EC 003.03	Manage CS Department Pre-Sail Checks		1 (review / debrief)		
EO 003.04	Support Towed Array Operations	2			
EC 003.04	Support Towed Array Operations		1		
EO 003.05	Contribute to the Project Management Process	10			
EC 003.05	Contribute to the Project Management Process		1		
PC 003	Coordinate Divisional Technical Readiness			1	
PO 006	**Advise Command on the CS Suite**	**66**	**18**	**6**	**90**
EO 006.01	Advise Command on the Communication Suite	20			
EC 006.01	Advise Command on the Communication Suite		3 (0.5 / trainee)		
EO 006.02	Advise Command on the Electronic Warfare Suite	2			

PO / EO	PO / EO Title	EO	EC*	PC*	PO
EC 006.02	Advise Command on the Electronic Warfare Suite		3 (0.5 / trainee)		
EO 006.03	Advise Command on the Navigation Suite	12			
EC 006.03	Advise Command on the Navigation Suite		3 (0.5 / trainee)		
EO 006.04	Advise Command on the Sonar Suite	7			
EC 006.04	Advise Command on the Sonar Suite		3 (0.5 / trainee)		
EO 006.05	Advise Command on the Fire Control Suite	15			
EC 006.05	Advise Command on the Fire Control Suite		3 (0.5 / trainee)		
EO 006.06	Advise Command on CS Suite Integration	10			
EC 006.06	Advise Command on CS Suite Integration		3 (0.5 / trainee)		
PC 006	Advise Command on CS Suite-Level Equipment			6 (1 / trainee)	
PO 001	**Coordinate Unit HAZMAT Responsibilities**	**12**	**6.5**	**6**	**24.5**
EO 001.01	Manage Unit HAZMAT Programme Readiness	3			
EC 001.01	Manage Unit HAZMAT Programme Readiness		0.5		
EO 001.02	Direct Response to a HAZMAT Incident	9			
EC 001.02	Direct Response to a HAZMAT Incident		6 (1/ trainee)		
PC 001	Coordinate Unit HAZMAT Responsibilities			6 (1 / trainee)	
PO 004	**Support Weapons Certification Programme (WCP)**	**53**	**16**	**7**	**76**
EO 004.01	Support WCP Administration	6			
EC 004.01	Support WCP Administration		1		
EO 004.02	Support WHDS Training	27			
EC 004.02	Support WHDS Training		9 (1.5 / trainee)		
EO 004.03	Support Weapons Employment for WCP	20			
EC 004.03	Support Weapons Employment for WCP		6 (1 / trainee)		

PO / EO	PO / EO Title	EO	EC*	PC*	PO
PC 004	Support Weapons Certification Programme (WCP)			7 (1 x TKT + 1 / trainee)	
PO 005	**Conduct the Duties of CS Engineering SME in HQ1**	**35**	**6**	**6**	**47**
EO 005.01	Sustain Combat Systems in Emergency Situations	35			
EC 005.01	Sustain Combat Systems in Emergency Situations		6 (1 / trainee)		
PC 005	Conduct the Duties of CS Engineering SME in HQ1			6 (1 / trainee)	
	Total	**226**	**53.5**	**29**	**308.5**

* Includes review and debrief time.

ANNEX H – ADDITIONAL TRAINING RESOURCES

IAW Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).

Estimates only, based on CF Cost Factors Manual Fiscal Year (FY) 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	3	$900.0	N/A	$2,700	$2,700	$13,500.00
R&Q	3	$36.0	45	$4,860	$4,860	$24,300.00
Incidentals	3	$17.5	45	$2,362.50	$2,362.50	$11,812.50
Subtotal				**$9,922.50**	**$9,922.50**	**$49,612.50**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	6	$50.0	1	$300	$300	$1,500.00
Classroom (square meters)	1	N/A (already in place)	45	N/A (already in place)	N/A	$0
Computer requirements	6	N/A (already in place)	1	N/A (already in place)	$0	$0
IT Equip (Proj, Smart board etc)	1	N/A	1	N/A	N/A	$0
DWAN connection (Set up)	1 room	N/A (already in place)	1	N/A (already in place)	$0	$0
Tools	N/A	N/A	N/A	N/A	N/A	N/A
Instructional Materials (Printed / Handouts etc.)	6	$30.0	1	$180	$180	$900.00
Subtotal				**$480.00**	**$480.00**	**$2,400.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)	N/A	N/A	N/A	N/A	N/A	N/A
		Subtotal		**$0**	**$0**	**$0**
	Total Estimated Costs			**$10,402.50**	**$10,402.50**	**$52,012.50**

- Max load – 6 (on the assumption of 3 trainees on TD and 3 local)
- TD costs to be significantly reduced with development of EOs recommended for TEL

Chap, 1, 2 approved 29 11 2016
Chap 3, 4 approved 15 03 2016
Version No 1.2
MITE Crse ID: 121048

ANNEX J – COURSE HISTORY

16 Jun 07

Creation of original course QSP.

QSP Change Request Version 1.01 08 Dec 2009

This QSP Change Request targeted all references in PO 001. Numerous students mentioned in their course critiques that having closed-book exams was not realistic with the day to day business of a working CSEC/O onboard submarine. Those comments were also brought forward in previous courses so it was decided by Submarine School to change all references to 'Will be provided' and 'are permitted' for all exams in PO 001.

QSPWB convened 18 Jan – 5 Feb 2016

The mandates of the board were to review and confirm the Master Task List, align AILN POs with AKPL, update systems/components to reflect those currently fitted, review/update HAZMAT training requirements, recommend POs/EOs for technology-enabled learning, and update/align the AILN QSP with NPTG QSP template. Furthermore, the board attempted to follow the Future Naval Training Strategy (FNTS) and strived to achieve a more practically based QSP/identify training for TEL. This course was previously 95 training days. A full review and reconstruction of the course occurred, and timings were updated based on historical values and SME input. Some jobs initially assumed to be the responsibility of the CSEC/O during the first iteration of this course were removed to actually reflect the job performed today.

QSP Change Request Ver 1.3 6 Aug 21

Change of equipment names in EO 004 due to MK48 Mod7AT weapon upgrade.

ANNEX K – ASSESSMENT PLAN

EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria
EC 001.01	EO 001.01	Manage Unit HAZMAT Programme Readiness	Written (self-study)	70%
EC 001.02	EO 001.02	Direct response to a HAZMAT incident	Practical	5/7 SAT
PC 001	EO 001.01-02	Coordinate Unit Hazardous Material Responsibilities	Practical	5/7 SAT
EC 002.01	EO 002.01	Manage CSED Safety IAW SEMS	Written	70%
EC 002.02	EO 002.02	Manage CSED Personnel Readiness	Written	70%
PC 002	EOs 002.01-02	Maintain Divisional Administrative Readiness	Practical	6/8 SAT
EC 003.01	EO 003.01	Manage Technical Administration	Written	70%
EC 003.02	EO 003.02	Coordinate unit-level Maintenance Activities	Written (self-study)	70%
EC 003.03	EO 003.03	Manage CS department pre-sail checks	Practical (self-study, scenario-based)	6/7 SAT
EC 003.04	EO 003.04	Support Towed Array Operations	Written (self-study)	70%
EC 003.05	EO 003.05	Contribute to the Project Management Process	Written (scenario-based)	70%
PC 003	EOs 003.01-04	Coordinate Divisional Technical Readiness	Practical	7/10 SAT + SAT on all critical performance factors
EC 004.01	EO 004.01	Support WCP Administration	Written	70%
EC 004.02	EO 004.02	Support WHDST	Practical	All SAT
EC 004.03	EO 004.03	Support Weapons Employment for WCP	Practical	14/17 SAT + SAT on all critical

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				performance factors
PC 004	EOs 004.01-03	Support WCP	Two part Practical (TKT + Trainer)	70% / All SAT
EC 005.01	EO 005.01	Sustain Combat Systems in Emergency Situations	Practical	All SAT (IAW SUBWOPS)
PC 005	EO 005.01	Conduct Duties of CS SME in HQ1	Practical	All SAT
EC 006.01	EO 006.01	Advise Command on the Communication Suite	Practical	5/7 SAT
EC 006.02	EO 006.02	Advise Command on the EW Suite	Practical	5/7 SAT
EC 006.03	EO 006.03	Advise Command on the Navigation Suite	Practical	5/7 SAT
EC 006.04	EO 006.04	Advise Command on the Sonar Suite	Practical	5/7 SAT
EC 006.05	EO 006.05	Advise Command on the Fire Control Suite	Practical	5/7 SAT
EC 006.06	EO 006.06	Advise Command on on CS Suite Integration	Practical	5/7 SAT
PC 006	EO 006.01-6	Advise Command on CS Suite	Practical	9/12 SAT

ANNEX L – DRAWING REQUIREMENTS

1. All sketches will be completed on Engineering Sketch Book (CF 1067);

2. There will be a written description including the applicable references put on a separate sheet of paper, which will be placed in front of the drawing;

3. Sketches shall be in block diagram format. Power, signal flow, and each individual environmental service will each be a single line colour-coded to a legend.

4. Sketches and lists shall be legible, logical, and accurate;

5. Power supplies, circuit designations and locations shall be shown;

6. Casualty power arrangements shall be shown, if applicable;

7. Systems shall be shown in as complete a manner as possible;

8. Ship's drawings maybe used for assistance. However, no photocopying will be permitted; and

9. Proper symbology and terminology must be used.

ANNEX M – COURSE DURATION CALCULATION FOR OPTIMAL/MINIMUM LOAD

	Periods	Minimal	Optimal
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	226	202.5	208
Enabling Checks (ECs)	53.5	15	30
Performance Checks (PCs)	29	9	17
Sub-total	**308.5**	**226.5**	**255**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5			
Base Administrative Time (BAT)	10	8	9
Training Administrative Time (TAT)	10	8	9
Training Travel Time (TTT)	0	0	0
Physical Fitness (PT)	24	18	21
Ceremonial/ Divisions	2	2	2
Sub-total	**46**	**36**	**41**
	354.5	**262.5**	**296**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) 8 (50 min)periods * **Partial days must be rounded up**	**354.5/8= 45 trg days**	**262.5/8= 33 trg days**	**296/8= 37 trg days**

***POs/EOs affected with optimal/minimum course load include all practical portions (EO 001.02, 004.02, 004.03, 005.01 and PO 006).**

ANNEX N - EO'S RECOMMENDED FOR TEL (LEARNING SUPPORT REQUESTS)

EO recommended for DL development	Duration of EO (periods)	Duration of EC/PC	Total duration/ residential trg time saved by developing DL content	DL content Requested (Ppt conversion, video, or simulation)	Justification for developing DL content	Security or controlled good issues? (Y/N)
EO 001.01	3	0.5	3.5	Ppt Conversion, Case Study to complete forms	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	n
EO 002.01	7	1	8 / Trg time saved - TBD	Ppt Conversion, Case Study to complete forms	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	n
EO 002.02	5	1	6 / Trg time saved - TBD	Ppt Conversion, Case Study to complete forms	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	n
PC 002		3	3 / Trg time saved - TBD	Case Study to complete forms	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	
PO 003	50	5	54 / Trg time saved - TBD	Ppt Conversion, Case Study to complete forms, Integrate with CSPS Risk/Project Management course	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	n

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EO 004.01	6	1	7 / Trg time saved - TBD	Ppt Conversion, Case Study to complete forms	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	n
PO006	66	ECs-18/ PC-6	90 / Trg time saved - TBD	Ppt Conversion, Interactive system block diagrams with electrical/signal flow simulation, interactive video/3D simulation	Lessens residential trg time / administrative trg which more effectively trained at mbr's home unit and cost as submarines training is just East coast; west coast trainees more time at home; continuation of training	Y (DL to contain only UNCLAS content). Secure discussions will occur during residential training
Total residential trg time saved			**171.5 / TBD**			